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Before the Proof

Claim Decomposition, Constructive Certification, and the Separation of Formal Result from Interpretive Meaning

A bidirectional architecture for problem-formation audit, primitive reconstruction, scoped verification, and interpretive attribution

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Scope. This paper develops a claim-routing and verification architecture that distinguishes empirical encounters from shareable registrations, registered observations, problem-forming assumptions, constructive primitives, formal or empirical adjudication, reconstructive accessibility, translation obligations, and downstream interpretation. It does not claim that every meaningful question is formally decidable, that every premise admits a unique decomposition, or that any verifier, canon, model, or institution supplies frame-independent final truth.

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Abstract

A proof, calculation, or empirical test can establish only what lies within its declared jurisdiction. Yet research arguments routinely move across several distinct operations: a target relation is registered through a measurement or representational system; the resulting datum is converted into a problem by choices of reference, identity, symmetry, ordering, and explanatory target; a formal or empirical traversal acts on that problem; and the result is translated into a broader physical, conceptual, or ontological interpretation. When these transitions are compressed, assumptions introduced at one layer can be misattributed to another.

This paper develops a bidirectional claim architecture for keeping those layers distinct while allowing them to cooperate. Its first direction decomposes a stated claim downward toward a declared primitive boundary. Its second direction constructs upward from that boundary through explicit grammars, finite construction objects, certificates, theorem-governed continuations, tests, and scoped completion conditions. The seam between the two directions is not asserted to contain metaphysically ultimate primitives. It is a declared stopping boundary for a particular receipt and can itself be reopened as a new claim.

The central registration chain is

$$X \xrightarrow{\mathcal{M}} D \xrightarrow{B_{\text{in}}} P \xrightarrow{F} Q \xrightarrow{B_{\text{out}}} I, \tag{1}$$

where X denotes the relation under interrogation, $\mathcal{M}$ a measurement or registration procedure, D the registered datum, B_{in} the problem-forming bridge, P the problem submitted for adjudication, F the formal, computational, or empirical traversal, Q the scoped result, and B_{out} the bridge to interpretation I. Verification of $F : P \to Q$ does not by itself verify either bridge.

The constructive counterpart is

$$\Pi = C^{(0)} \xrightarrow{\Gamma} C^{(1)} \to C^{(2)} \to \cdots \to C^{(n)} \xrightarrow{\mathsf{Ver}} \mathfrak{R}, \tag{2}$$

where Π is a declared primitive set, Γ an admissible grammar, $C^{(k)}$ traceable construction objects, Ver an appropriate verifier, and $\mathfrak{R}$ a scoped receipt. Existing IRRPROOF, Relational Crucible, finite-witness/infinite-registry, positional-recoverability, DST, and QED-completion work provides in-corpus instantiations of this constructive side, not independent replication: finite calculation is distinguished from theorem-governed continuation; negative reachability certificates are distinguished from failed searches; machine-checkable construction packages retain dependencies and residue; and completion is defined relative to an explicit contract rather than a frame-independent final truth.

The method introduces typed Claim Kernels, Problem-Formation Audits, Bridge Attribution Error, Reference Reification Error, Zero-Type Error, a Declared Primitive Stop Rule, Reconstructive Accessibility, and Scoped Verification Receipts. Two physical exemplars show why the separation matters. Baryon asymmetry demonstrates upstream problem formation: a registered empirical estimate can support different declared explanatory problems. Quantum-vacuum and zero-point-energy language demonstrates downstream type migration: exact statements about ground states, expectation values, variances, and energy differences can acquire additional semantic content through unmarked transitions among *zero*, *vacuum*, *null*, *empty*, and *nothing*.

The proposal does not demote convention, standardization, established theory, or expert judgment. It asks instead that formal validity, empirical adequacy, reconstructive understanding, and interpretive assent remain independently attributable. The practical objective is a neutral routing layer in which mechanically iterable claims can be sent to formal engines while linguistic, philosophical, and correspondence questions remain separately visible and auditable. A further distinction developed here is prior to the usual empirical/formal split. An empirical encounter is not itself transferred between observers. What can be shared is a registration produced from an encounter, together with its procedure, provenance, calibration, transformation history, and comparison rule. Formal verification therefore acts on registered descriptions, not on another observer's originating experience. The same discipline applies to variables, zeros, objects, causal arrows, and reality predicates: a successful representation may parameterize, discriminate, or close an accounting without thereby acquiring a stronger physical, causal, or ontological status. The framework consequently adopts paired non-promotion and non-demotion rules: do not

infer a stronger status than the registered warrant supports, and do not infer falsity merely because a stronger status has not been established.

The framework is also self-applied to Proposition 3.2 through a filled Claim Kernel, a finite typed decomposition graph, a declared stopping boundary, a reconstruction test, an adequacy determination, and a scoped verification receipt; the resulting adequacy judgment remains persistently challengeable rather than self-ratifying.

The companion paper *Lambda as Complementary Modeling* functions here as a worked case of the broader method: it holds an empirical target fixed while comparing what changes when registry commitments and interpretive bridges are made explicit.

1. Introduction: Verification Has Jurisdiction

A contemporary research paper may contain a theorem, a numerical calculation, a measurement, a model identification, a causal explanation, a semantic equivalence, and an ontological conclusion in a few adjacent paragraphs. Those statements do not all have the same logical type, and they cannot all be certified by the same procedure.

A theorem prover can check a formal consequence of declared axioms. It cannot, merely by checking that consequence, establish that a symbol in the theorem corresponds to a particular physical object. An experiment can constrain an observable under a specified protocol. It cannot, merely by returning a null result, witness ontological absence. A successful physical model can explain a phenomenon relative to its primitives and correspondence rules. Its success does not automatically establish that those primitives form the uniquely necessary way to formulate the phenomenon.

This paper treats these distinctions as an engineering problem in knowledge representation rather than as a demand to settle every philosophy-of-science dispute. The goal is to expose the transitions that already occur in ordinary reasoning and to attach a scoped receipt to each transition that can actually be checked.

The method begins from a simple observation:

$$\boxed{\text{datum} \neq \text{problem} \neq \text{formal result} \neq \text{interpretation.}} \tag{3}$$

These objects can be related without being identical.

A second observation follows. A problem does not arise from a datum alone. It arises from a datum together with a registry of definitions, references, equivalence relations, admissible transformations, calibration procedures, ordering choices, and explanatory expectations. Consequently, two investigators may agree completely about a datum and nevertheless formulate different problems from it.

A third observation concerns the other end of an argument. A formal result may support an interpretation, but the correspondence between the formal object and the interpretation can contain assumptions not proved by the formal derivation itself.

The proposed filter therefore audits both sides of formal adjudication.

2. Scope and Non-Exclusivity

The method is complementary. It does not propose replacing ordinary proof, peer review, experiment, dimensional analysis, statistical inference, theorem proving, or disciplinary expertise. It attempts to prepare claims for those systems and to prevent a receipt from one system from silently acquiring jurisdiction over another claim type.

No claim is made that every meaningful statement can be formalized. No claim is made that interpretive or philosophical questions become meaningless when they resist formalization. The opposite discipline is intended: a question should not be rejected merely because a formal adjudicator lacks jurisdiction over it, and a formal adjudicator should not be credited with deciding a question that was never represented in its input.

The method also avoids the converse error that convention implies arbitrariness. Standardized conventions often survive precisely because they are reproducible, interoperable, and robust under sustained application. Their conventional history does not diminish those properties. What remains open to audit is the further inference from

$$\text{robust in a declared domain} \tag{4}$$

to

$$\text{uniquely necessary description of the target.} \tag{5}$$

The latter requires an additional argument.

2.1 Global meta-rule: scoped assessment, inspectable residue, and non-zeroing

The framework applies the following rule to every registered claim, observation, inference, challenge, ambiguity, obstruction, classification, scope declaration, and adjudicative finding, including statements made by the framework itself.

Scoped Assessment and Non-Zeroing Meta-Rule. Every registered object is assessed relative to its declared scope, assumptions, dependencies, jurisdiction, resolution, and intended use. Registration of a challenge establishes that the challenge has been made and identifies its target; registration alone neither establishes the challenge as dispositive nor requires its elimination. In particular, $\mathrm{Detect}(q) \not\Rightarrow \mathrm{MustResolve}(q)$. A finding blocks a traversal only when it materially prevents satisfaction of a requirement already declared for that traversal. A detected but non-material finding therefore does not, by detection alone, imply traversal failure.

This rule does not weaken discovery. An auditor should identify every technically supportable obstruction, ambiguity, alternative interpretation, dependency, hidden assumption, bridge problem, or contradiction encountered. Discovery and disposition are separate operations. A registered finding may be corrected, blocking, contested, an accepted limitation, unresolved, externally adjudicated, outside the present jurisdiction, or non-blocking under the declared scope. Disposition does not imply elimination.

Formalization Boundary. The framework does not require every meaningful statement, objection, ambiguity, interpretation, or scope dispute to admit mechanical resolution. Its purpose is to identify relations that can be formalized and route those relations to formal verification without silently extending the resulting verdict to claims the formal procedure did not establish. Residual interpretive questions remain inspectable registry objects rather than being coerced into artificial formal closure. A machine-verifiable receipt therefore certifies only the formal object, rule set, premises, dependencies, and scope actually submitted to that verification procedure.

Persistent Challengeability and Novelty Rule. No certificate, receipt, verdict, stopping declaration, adequacy determination, or prior audit disposition makes a claim unchallengeable. A challenge may be registered at any later stage. Registration neither establishes the challenge nor defeats the challenged claim. Each challenge retains its target, asserted obstruction, relevant premises and scope, and its relation to prior registered challenges. A materially equivalent restatement may be linked to an existing challenge without being counted as a novel challenge; a materially distinct obstruction, dependency, counterexample, premise dispute, or scope conflict remains separately inspectable. Equivalence here records challenge provenance, not dismissal: a repeated challenge does not cease to matter merely because it was previously registered. Novelty and blocking status remain independent. Certification records successful traversal of a declared verification contract; it does not certify semantic finality outside that contract. Repeated audit therefore seeks convergence in the registry of materially novel known challenges, not the impossible elimination of every conceivable objection.

Contribution-floor rule. The paper's own abstract-level contribution claims are registered requirements of this traversal. A scope restriction that preserves a local result while defeating a contribution claimed by the paper is therefore challengeable at the contribution level and may be blocking there. This supplies a declared floor for scope without granting any auditor unrestricted jurisdiction: the relevant question remains whether the challenged scope can support the contribution actually claimed.

Scope itself remains inspectable. A restriction of scope, jurisdiction, resolution, or admissible assumptions that changes the disposition of a finding is itself a registered commitment and may itself be challenged. Thus neither an author nor an auditor may obtain immunity or unrestricted jurisdiction merely by declaring an issue material or immaterial.

Successful traversal consequently does not require all registered residue to vanish. Nonzero residue may remain provided it is typed, localized, attributable, and inspectable. Readiness instead requires that every known finding material to the declared requirements possess an inspectable disposition and that no undisposed blocking finding prevent satisfaction of those requirements. The registry administers these relations; it does not convert registration into a ruling about frame-independent truth.

Auditor application. Exhaustive discovery and mandatory correction shall not be conflated. Before classifying a finding as traversal-blocking, an auditor shall identify its target, the declared requirement affected, the relevant scope or jurisdiction, and how the finding prevents that requirement from being satisfied. A finding for which this blocking relation has not been established remains reportable and inspectable without becoming blocking merely because it remains unresolved. Different auditors may assign different significance to the same registered challenge; those assessments remain attributable registered objects rather than silently collapsing into a single verdict.

3. The Five-Stage Chain

Definition 3.1 — Registration chain

Let

$$\mathsf{Ch} = (X, \mathcal{M}, D, B_{\mathrm{in}}, P, F, Q, B_{\mathrm{out}}, I). \tag{6}$$

Its components are:

- X: the physical, mathematical, or conceptual relation being interrogated;
- $\mathcal{M}$: the registration or measurement registry;
- D: the resulting registered datum or stipulated input;
- B_{in}: the bridge that transforms D into a problem;
- P: the problem actually submitted for adjudication;
- F: the formal, computational, empirical, or mixed traversal;
- Q: the result licensed by that traversal;
- B_{out}: the interpretive or correspondence bridge;
- I: the broader conclusion.

The chain is

$$X \xrightarrow{\mathcal{M}} D \xrightarrow{B_{\mathrm{in}}} P \xrightarrow{F} Q \xrightarrow{B_{\mathrm{out}}} I. \tag{7}$$

The registration-chain schema is descriptive, not metaphysical. It does not assert that reality literally proceeds through these stages. It describes the information dependencies of an argument.

Proposition 3.2 — Bridge-underdetermination lemma

Let two registration chains share the same formal problem, traversal, and result,

$$\mathsf{Ch}_1 = (X_1, \mathcal{M}_1, D_1, B^{(1)}_{\mathrm{in}}, P, F, Q, B^{(1)}_{\mathrm{out}}, I_1), \tag{8}$$

$$\mathsf{Ch}_2 = (X_2, \mathcal{M}_2, D_2, B^{(2)}_{\mathrm{in}}, P, F, Q, B^{(2)}_{\mathrm{out}}, I_2). \tag{9}$$

Assume that the scoped verification relation for F depends only on $(P, \mathsf{Prem}_F, \Gamma_F)$ and yields the same accepted result Q in both chains. Suppose further that $B^{(1)}_{\mathrm{in}}$ satisfies a declared adequacy criterion A_{in} while $B^{(2)}_{\mathrm{in}}$ does not. Then acceptance of $P \xrightarrow{F} Q$ cannot distinguish the adequacy of the two upstream bridges.

Proof. By construction, the verifier receives the same formal problem P, premise boundary Prem_F, and grammar Γ_F in both chains, and therefore returns the same scoped result Q. Yet the two chains differ in a property of the upstream bridge not represented in the verifier input:

$$A_{\mathrm{in}}(B^{(1)}_{\mathrm{in}}) = 1, \qquad A_{\mathrm{in}}(B^{(2)}_{\mathrm{in}}) = 0. \tag{10}$$

Hence the formal acceptance of $P \xrightarrow{F} Q$ underdetermines upstream bridge adequacy. The same construction applies mutatis mutandis to a downstream interpretive bridge. □

The result is deliberately modest. It does not say that bridges are unknowable or uncheckable. It says only that a verifier cannot certify assumptions or correspondences that are absent from its declared input and jurisdiction.

4. Measurement Registries

A measurement does not map an uninterpreted world directly to a naked number. A reproducible measurement requires a specified coupling, comparator, calibration, enumeration procedure, uncertainty model, and representation.

Definition 4.1 — Measurement registry

A measurement registry is a tuple

$$\mathcal{M} = (\chi, \Delta, \mathcal{E}_{\text{enum}}, U_{\text{cal}}, \mathcal{N}, \epsilon), \tag{11}$$

where χ specifies the coupling or observable, Δ a selected increment or comparison structure, $\mathcal{E}_{\text{enum}}$ an enumeration or estimation procedure, U_{cal} the unit/calibration convention, $\mathcal{N}$ the numerical representation, and ϵ the resolution or uncertainty structure.

The registered value is therefore written

$$D = q_{\mathcal{M}}(X), \tag{12}$$

rather than treating the numerical report as identical to X.

This does not make measurement subjective. Standardizing $\mathcal{M}$ is what allows independent observers to reproduce a comparison.

4.1 The cesium second as an exemplar

The SI second illustrates the distinction. Modern SI fixes the numerical value of the unperturbed ground-state hyperfine transition frequency of the cesium-133 atom. The physical transition, the selected transition frequency, the exact assigned numerical value, the unit second, and a measured frequency in hertz participate in one calibrated registry. They should not be collapsed into one ontological object.

The important methodological point is not a criticism of SI. It is that standardized measurement exposes a declared division/enumeration structure:

$$\begin{gathered}\text{reproducible phenomenon} \to \text{selected transition} \to \text{counted periodic relation} \\ \to \text{defined unit} \to \text{reported frequency.}\end{gathered} \tag{13}$$

Once standardized, this infrastructure becomes intentionally transparent in ordinary use. The filter restores it only when a claim depends on the difference between the physical relation, the increment, the unit, or an ideal limit.

4.2 Empirical encounter, registration, and intersubjective comparison

The framework distinguishes an empirical encounter from the shareable record produced from it. Let $\mathcal{E}_O$ denote an encounter involving observer or apparatus O, and let

$$\mathcal{E}_O \xrightarrow{\mathcal{M}_O} d_O \tag{14}$$

be the registration that produces a transportable record d_O. Another observer does not receive $\mathcal{E}_O$; that observer can receive, inspect, copy, or transform d_O, and can perform a new procedure that produces a new encounter and a new record.

Replication therefore has the form

$$\mathcal{E}_{O_i} \xrightarrow{\mathcal{M}_i} d_i, \qquad d_i \sim_{\varepsilon,\Gamma} d_j, \tag{15}$$

where the equivalence criterion, tolerance, and comparison grammar are declared. This is not skepticism about experiment. It is a more exact account of what experiment makes public: procedure, registration, provenance, calibration, transformation history, and reproducible correspondence among independently produced records.

Accordingly, the routing tag E in this paper should be read as a registration/empirical-report/statistical claim type. A machine can check a stored value, calibration relation, statistical statement, or provenance chain when those objects are available to it. It does not thereby re-experience the originating empirical encounter.

4.3 Variables as declared differentiation

Introducing a variable already declares a state domain and an admissible direction of differentiation. For a registry R, write

$$D_R : X \to \mathcal{L}_R, \tag{16}$$

where $\mathcal{L}_R$ is the set of distinctions retained by that registry. The induced equivalence relation is

$$x \sim_R y \iff D_R(x) = D_R(y). \tag{17}$$

Thus $x \neq y$ can coexist with $D_R(x) = D_R(y)$. Equality in the registered image does not, by itself, establish identity in the source.

For a cyclic degree of freedom one may write

$$\theta \in S^1, \qquad \theta \sim \theta + 2\pi, \tag{18}$$

and a finite detector may retain only a quotient

$$\kappa_N : S^1 \to \{0, 1, \dots, N-1\}. \tag{19}$$

The N reported states are exhaustive in the codomain of κ_N; that fact alone does not prove that the source consists of only N primitive states. In the binary case, exhaustive output in $\{0, 1\}$ therefore does not entail exhaustive ontology in the source domain.

This motivates a registry-level definition: a degree of freedom is an independently admissible direction of differentiation in a declared state domain; a variable parameterizes that differentiation. A variable becomes operationally physical only through a declared correspondence to possible registrations or model-mediated inferences. The predicates "physical," "measurable," "observable," and "real" are therefore not treated as synonyms. A reality predicate must expose its prerequisites; schematically,

$$\mathrm{Real}_R(x) \tag{20}$$

means only that x satisfies the criterion for reality declared by registry R, not that every admissible registry must use the same criterion.

4.4 From astronomical recurrence to the second and musical pitch

The history of timekeeping makes the distinction concrete. A day is not introduced into mathematics as a perfectly noiseless primitive period. Astronomical recurrence supplies a physically varying relation from which communities construct stabilized time scales and subdivisions. The familiar civil partition

$$1 \text{ day} = 24 \text{ hours} = 24 \cdot 60 \cdot 60 \text{ seconds} \tag{21}$$

contains a division structure before any decimal numeral is written. The decimal representation of 86400 is another representational choice. Modern SI then defines the second through a fixed cesium-133 transition-frequency value rather than by treating a particular astronomical day as an exact duration. The lesson is not that the standard is defective; it is that an exact standard can be constructed precisely because variable physical recurrences are not themselves exact standards.

A musical scale makes the same layering audible. In twelve-tone equal temperament, relative frequencies obey

$$f_n = f_0\, 2^{n/12}. \tag{22}$$

Once f_0, the time unit, and the tuning convention are declared, the relation is exact inside that registry. The exactness of the formula does not make the selected pitch reference, twelvefold octave partition, note names, or underlying time standard uniquely necessary features of sound. Standardization creates a reproducible accounting surface; it does not erase the provenance of the divisions that make the numerals meaningful.

5. Claim Kernels

A document is not the atomic object of verification. Claims are.

Definition 5.1 — Claim Kernel

For a claim C, define

$$\mathcal{K}(C) = (C, T, \mathsf{Prem}, \mathsf{Def}, \Gamma, J, \mathcal{X}, \rho), \tag{23}$$

where:

- T is the provisional claim type;
- Prem is the declared premise set;
- Def contains definitions and typed dependencies;
- Γ contains admissible inference or transformation rules;
- J specifies adjudication jurisdiction;
- $\mathcal{X}$ contains bridges to other kernels;
- ρ records unresolved residue.

Let the provisional claim-type set be $\mathcal{T}$, with the field T taking values in that set:

$$\mathcal{T} := \{F, E, S, M, B, Z\}, \qquad T \in \mathcal{T}. \tag{24}$$

Here F denotes formal or mathematical claims; E, empirical or statistical claims; S, semantic or interpretive claims; M, methodological, measurement, or conventional claims; B, bridge or correspondence claims; and Z, reference, zero, or limit claims. These are routing labels, not mutually exclusive ontological categories. A mixed claim should normally decompose into several kernels.

Definition 5.2 — Adequate kernel decomposition

Let $G_C = (V_C, E_C)$ be a nonempty finite directed acyclic graph of Claim Kernels proposed as a decomposition of scoped claim C, with distinguished root $\mathcal{K}(C) \in V_C$. Each vertex therefore has the type $\mathcal{K}(C_i)$ for some scoped subclaim, premise-bearing claim, or bridge claim C_i, rather than being an untyped bare claim. Every edge is oriented downward from a kernel toward a typed subclaim, premise-bearing kernel, or bridge obligation on which reconstruction of the source kernel depends. The root has in-degree zero, and every other vertex is reachable from $\mathcal{K}(C)$ by a directed path. Thus every vertex and every leaf belongs to the rooted decomposition; disconnected stopped components are inadmissible. Finiteness, acyclicity, rootedness, reachability, and this edge orientation are part of the admissible graph type.

Define the leaf boundary by out-degree zero:

$$L_C := \mathrm{Leaves}(G_C) = \{\mathcal{K}(C_i) \in V_C : \mathrm{outdeg}_{G_C}(\mathcal{K}(C_i)) = 0\}. \tag{25}$$

Because G_C is a nonempty finite rooted DAG, $L_C \neq \emptyset$.

Let

$$\mathrm{Rec}_{\Gamma_C}(L_C; E_C) \tag{26}$$

denote claims reconstructible from the declared leaf boundary by reverse traversal of the recorded edge structure E_C under the typed reconstruction obligations declared by Γ_C. Non-leaf vertices are intermediate reconstruction targets, not additional starting premises. No unrecorded recovery route may satisfy completeness. Each edge type may license a downward decomposition step while carrying a corresponding upward reconstruction obligation; Sound and Complete test these two directions independently. An edge's admission for decomposition therefore does not silently assert that its reverse is deductively valid unless the declared reconstruction obligation supplies that relation.

Soundness constrains the decomposition edges themselves:

$$\mathrm{Sound}(G_C; C) \iff \text{every edge in } E_C \text{ is licensed by the declared typed decomposition or bridge calculus } \Gamma_C. \tag{27}$$

Completeness constrains reconstruction in the opposite direction:

$$\mathrm{Complete}(G_C; C) \iff C \in \mathrm{Rec}_{\Gamma_C}(L_C; E_C), \tag{28}$$

where reconstruction must proceed through the recorded edge structure E_C and the reconstruction obligations declared by Γ_C as specified above. Thus downward decomposition must be licensed edge by edge, while the root must be recoverable upward from the declared stopping boundary. Calibration assumptions, instrument records, semantic premises, and bridge obligations need not be deductive consequences of C merely to serve as typed grounds.

A decomposition is **proper relative to the declared stopping boundary** when every leaf kernel in L_C satisfies a stopping predicate selected by a declared leaf-stopping assignment $\mathsf{LeafStop}_C$. For each leaf, $\mathsf{LeafStop}_C$ records the applicable jurisdiction J and whether stopping is effective or declared; when stopping is declared, it also records the responsible actor H. Definition 10.1 supplies those jurisdiction- and actor-relative truth conditions.

Accordingly, the one-vertex identity graph $V_C = \{\mathcal{K}(C)\}, E_C = \emptyset$ is admissible under Proper only when the root kernel itself satisfies the selected stopping predicate. In the declared-stop branch this is explicitly an actor-relative boundary declaration, not an independent proof that the root is primitive; the actor H and boundary statement remain on the receipt. Because $\mathsf{LeafStop}_C$ is declared metadata of G_C, it need not be added as a separate argument each time Proper or Adeq is written. Write

$$\mathrm{Proper}(G_C; C). \tag{29}$$

An **admissible decomposition** satisfies all three conditions:

$$\mathrm{Adeq}(G_C; C) := \mathrm{Sound}(G_C; C) \land \mathrm{Complete}(G_C; C) \land \mathrm{Proper}(G_C; C). \tag{30}$$

The calculi remain typed. A partly empirical, semantic, methodological, or bridge-bearing decomposition must use the corresponding licensed rule type rather than disguise that dependency as formal deduction. Adeq certifies only the structural conditions explicitly assigned to it. It does not promote every registered grounding commitment to the epistemic status of the root claim, determine the ultimate necessity of every premise or bridge, or require all residual challenges to vanish. Surplus commitments and unresolved distinctions remain registered, attributable, and open to challenge under the global Scoped Assessment and Non-Zeroing Meta-Rule.

An adequacy result is therefore informative only together with its recorded leaf-stopping profile. Any presentation of $\mathrm{Adeq}(G_C; C) = 1$ shall surface which leaves stopped effectively and which stopped by declaration; for every declared stop it shall also surface the responsible actor H. Thus structural adequacy does not erase the warrant distinction already encoded by $\mathsf{LeafStop}_C$.

A useful decomposition need not be unique. Two auditors may produce $G_C^{(1)} \neq G_C^{(2)}$ while both satisfy Adeq. The disagreement then becomes a comparable registered object rather than an undefined appeal to preserving inferential content.

Section 12 supplies a concrete witness to the registry-dependent problem-formation structure developed in Section 6: one registered empirical estimate is held fixed while distinct declared problem-forming bridges generate different submitted problems.

6. Problem-Formation Audits

The central upstream question is not merely *Is the proposed solution correct?* It is also *What transformed the observation into this particular problem?*

Let $\mathcal{G}$ denote a problem-generating registry. Then

$$P_{\mathcal{G}}(D) = \Phi(D; \mathcal{G}). \tag{31}$$

Two registries can satisfy

$$D_{\mathcal{G}_1} = D_{\mathcal{G}_2} = D \tag{32}$$

while

$$P_{\mathcal{G}_1}(D) \neq P_{\mathcal{G}_2}(D). \tag{33}$$

The registry-dependent problem inequality does not imply that either problem is illegitimate. It means the problem statement contains structure beyond the datum.

Definition 6.1 — Problem-Formation Kernel

Write

$$B_{\text{in}} = (\mathsf{Prim}, \mathcal{E}_{\text{eqv}}, \mathcal{C}_{\text{corr}}, \mathcal{O}_{\text{ord}}, \mathcal{S}_{\text{sym}}, \mathcal{Z}), \tag{34}$$

where:

- Prim: mathematical or logical primitives;
- $\mathcal{E}_{\text{eqv}}$: identity and equivalence conventions;
- $\mathcal{C}_{\text{corr}}$: calibration and correspondence assumptions;
- $\mathcal{O}_{\text{ord}}$: ordering, orientation, or causal assumptions;
- $\mathcal{S}_{\text{sym}}$: symmetry or invariance commitments;
- $\mathcal{Z}$: reference states, nulls, ideals, or limiting values.

A Problem-Formation Audit asks which of these components are observed, stipulated, inherited from disciplinary convention, or introduced for explanatory convenience.

Definition 6.2 — Upstream Problem Attribution Error

An upstream problem attribution error occurs when structure introduced by B_{in} is attributed to D as though it were contained in the datum itself.

The pattern is

$$D \xrightarrow{B_{\text{in}}} P \quad \rightsquigarrow \quad \text{“the datum itself demands } P\text{.”} \tag{35}$$

The error is not the use of a bridge. The error is failure to account for it.

7. Formal Jurisdiction

For a formal traversal,

$$F : (P, \mathsf{Prem}_F, \Gamma_F) \mapsto Q, \tag{36}$$

the verification target must identify the language, premise set, inference rules, and stopping conditions.

A machine-checkable proof provides an especially clean receipt because its formal jurisdiction is explicit. Numerical simulation, symbolic algebra, statistical inference, and empirical testing can be treated analogously, although their assumptions and error models differ.

Definition 7.1 — Formal receipt

A formal receipt has the form

$$\mathfrak{R}_F = (\mathcal{K}, J_F, A_F, K_F^{\checkmark}, \rho_F), \tag{37}$$

where A_F records accepted premises, $K_F^{\checkmark}$ the verified content, and ρ_F unresolved or out-of-jurisdiction content.

The important rule is:

$$\boxed{\text{receipts attach to scoped claims, not indiscriminately to documents.}} \tag{38}$$

A paper containing one proved theorem and five interpretive claims does not become a proved paper.

8. Interpretive Bridges

A result Q often becomes scientifically useful only after correspondence with a broader target. The method does not oppose this step. It makes the step explicit.

Let

$$B_{\text{out}} : Q \mapsto I. \tag{39}$$

The bridge may contain physical identification, semantic equivalence, causal interpretation, ontological commitment, approximation, extrapolation, or explanatory criteria.

8.1 Bridge attribution and jurisdiction

Definition 8.1 — Bridge Attribution Error. A Bridge Attribution Error (BAE) occurs when content required by B_{out} is attributed to Q as though it had been established by the adjudication of Q.

Thus

$$F \vdash Q \not\Rightarrow F \vdash I(Q) \tag{40}$$

unless the bridge premises are included:

$$F + B_{\text{out}} \vdash I(Q). \tag{41}$$

This is a type-checking discipline for inference. It does not prejudge whether I is true.

8.2 Non-promotion, non-demotion, and causal arrows

A successful downstream calculation does not retroactively strengthen the epistemic type of its inputs. The framework therefore applies two paired rules:

$$\text{do not promote beyond warrant,} \qquad \text{do not demote beyond warrant.} \tag{42}$$

A model-introduced parameter is not thereby a discovered entity; failure to establish it as an entity does not thereby prove it fictitious. A correlation is not thereby a cause; failure to establish a causal bridge does not thereby prove that no causal relation exists.

The same discipline applies to arrows. The notation $X \to Y$ may encode functional dependence, temporal order, intervention response, conditional probability, coupling, derivation, or an asserted causal mechanism. Those meanings are not interchangeable. Whenever a causal interpretation matters to the conclusion, the meaning of the arrow and the warrant for that meaning belong in the bridge registry rather than being inherited from the glyph.

9. References, Ideals, and the Zero Audit

The token 0 carries heterogeneous mathematical and linguistic roles. A single occurrence may have more than one legitimate classification. The audit therefore separates what a zero **denotes** from what role it **plays** in a registry.

Definition 9.1 — Zero denotation and role

For a substantive occurrence z of a zero-like token, define

$$\mathsf{ZeroClass}(z) = (Z_{\text{den}}(z), Z_{\text{role}}(z)), \tag{43}$$

where each component is a finite subset of a declared vocabulary.

A minimal denotation vocabulary is the finite label set

$$\mathcal{Z}_{\text{den}} = \{z_{\text{alg}}, z_{\text{lim}}, z_{\text{state}}, z_{\text{meas}}, z_{\text{ont}}\}, \qquad Z_{\text{den}}(z) \subseteq \mathcal{Z}_{\text{den}}. \tag{44}$$

The labels denote, respectively: algebraic identity; limit value; state or excitation label; measured estimate; and ontological absence claim.

A minimal role vocabulary is

$$\mathcal{Z}_{\text{role}} = \{z_{\text{orig}}, z_{\text{ref}}, z_{\text{cal}}, z_{\text{ren}}, z_{\text{sym}}, z_{\text{null}}\}, \qquad Z_{\text{role}}(z) \subseteq \mathcal{Z}_{\text{role}}. \tag{45}$$

The labels denote, respectively: coordinate origin; reference level; calibration baseline; renormalization convention; symmetry or equilibrium ideal; and experimental null threshold.

For a scoped audit, let $\mathcal{O}_{\text{act}}$ denote the declared finite set of inferentially active zero-like occurrences to be classified. The classifier applies $\mathsf{ZeroClass}(z)$ to each $z \in \mathcal{O}_{\text{act}}$. Semantic admission is intentionally not manufactured by the formal minimizer. Before minimization, a declared actor supplies a finite adjudication domain $\mathcal{D}_S \subseteq \mathcal{O}_{\text{act}} \times (\mathcal{Z}_{\text{den}} \cup \mathcal{Z}_{\text{role}})$ and records an explicit admitted/rejected judgment for every pair in $\mathcal{D}_S$. Write the admitted relation on that declared domain as $\text{Admissible}_S(z, \ell)$. Unadjudicated occurrence-label pairs lie outside $\mathcal{D}_S$; they are not silently treated as false. The actor, domain, and judgments are retained as inspectable type-S residue.

A candidate accounting pair satisfies $\widehat{\mathcal{Z}}_{\text{den}} \subseteq \mathcal{Z}_{\text{den}}$ and $\widehat{\mathcal{Z}}_{\text{role}} \subseteq \mathcal{Z}_{\text{role}}$. Such a pair is *$S$-admissible for $\mathcal{O}_{\text{act}}$* when, for every $z \in \mathcal{O}_{\text{act}}$, it contains at least one denotation label ℓ_d and at least one role label ℓ_r such that $(z, \ell_d), (z, \ell_r) \in \mathcal{D}_S$ and both $\text{Admissible}_S(z, \ell_d)$ and $\text{Admissible}_S(z, \ell_r)$ are registered as admitted. This pair-level predicate is therefore defined entirely relative to the finite semantic adjudication already supplied; it does not require a decision over unadjudicated labels.

The **minimal-cover rule** is then formal relative to that registered semantic input. An S-admissible accounting pair $(\widehat{\mathcal{Z}}_{\text{den}}, \widehat{\mathcal{Z}}_{\text{role}})$ is *jointly inclusion-minimal* when there is no other S-admissible pair $(\widehat{\mathcal{Z}}'_{\text{den}}, \widehat{\mathcal{Z}}'_{\text{role}})$ such that $\widehat{\mathcal{Z}}'_{\text{den}} \subseteq \widehat{\mathcal{Z}}_{\text{den}}$ and $\widehat{\mathcal{Z}}'_{\text{role}} \subseteq \widehat{\mathcal{Z}}_{\text{role}}$, with at least one inclusion strict. This is minimality in the product order, so coordinated reductions across the two axes are tested rather than holding one axis fixed while minimizing the other. Formal minimization therefore certifies minimality relative to the registered semantic judgments; it does not retroactively machine-certify those judgments. A calibration baseline can also denote the algebraic identity; a renormalization zero can be a special case of a reference zero. Multiple membership is permitted precisely so that these relations need not be hidden. Minimal accounting pairs need not be unique; when two or more incomparable jointly minimal pairs remain admissible, the audit records the alternatives rather than imposing an undeclared tie-break.

No cross-category identity is presumed merely because the same glyph is used.

9.1 Experimental nulls are not sensed absence

A detector returns a physical state or record. A null result normally has a form such as

$$|\hat{x} - x_0| < \epsilon_{\mathcal{M}}, \tag{46}$$

or, statistically,

$$H_0 : x = x_0 \quad \text{is not rejected under the declared model and sensitivity.} \tag{47}$$

Neither statement is identical to an ontological absence claim.

Finite measurement can approach increasingly restrictive bounds,

$$\epsilon_1 > \epsilon_2 > \epsilon_3 > \cdots \to 0, \tag{48}$$

without any finite station becoming sensory access to the limit itself.

Definition 9.2 — Zero-Type Error

A Zero-Type Error occurs when an argument moves from one declared denotation or role in $\mathsf{ZeroClass}(z)$ to another without an explicit bridge sufficient for the conclusion. Examples include treating a state label $|0\rangle$ as zero physical content, treating a zero expectation value as pointwise absence, or treating an experimental null as proof of nonexistence.

Definition 9.3 — Reference Reification Error

Fix a common registered quantity D under a measurement registry whose coupling χ, unit/calibration convention U_{cal}, and numerical representation $\mathcal{N}$ are held fixed. Let two comparison registries differ only in their selected reference values $z_{\mathcal{G}}$ and $z_{\mathcal{H}}$. Define

$$\Delta_{\mathcal{G}}(D) = D - z_{\mathcal{G}}, \qquad \Delta_{\mathcal{H}}(D) = D - z_{\mathcal{H}}. \tag{49}$$

Then

$$\Delta_{\mathcal{H}}(D) - \Delta_{\mathcal{G}}(D) = z_{\mathcal{G}} - z_{\mathcal{H}}. \tag{50}$$

A Reference Reification Error occurs when the privileged status of one selected reference is attributed to the registered quantity itself rather than to the declared comparison registry.

The common-carrier hypothesis matters. If the measurement coupling, unit calibration, or representation also changes, then a translation map must first establish that the two registered quantities are comparable.

10. Recursive Decomposition and the Primitive Stop Rule

If every bridge can become another claim, decomposition can recurse:

$$C \to \{C_1, C_2, B_1\} \to \{\mathcal{K}(C_1), \mathcal{K}(C_2), \mathcal{K}(B_1)\} \to \cdots. \tag{51}$$

This self-propagating behavior is useful only if local adjudication can terminate.

Definition 10.1 — Declared Primitive Stop Rule

Let A_J be the accepted boundary for jurisdiction J.

The effective-stop predicate is evaluated only when its membership test is operationally defined: require the kernel premise set $\mathsf{Prem}_{\mathcal{K}}$ to be finite and require membership in A_J to be decidable (for example because A_J is finite or is given by a finite decidable grammar). Under those preconditions define

$$\mathrm{Stop}_J^{\text{eff}}(\mathcal{K}) \iff \mathsf{Prem}_{\mathcal{K}} \subseteq A_J. \tag{52}$$

If those effective-stop preconditions are unavailable, stopping may instead be a **declared human or external judgment**:

$$\mathrm{Stop}_J^{\text{decl}}(\mathcal{K}; H) = 1, \tag{53}$$

and the receipt must identify the actor H, the boundary statement, and any unresolved membership questions.

Stopping does not declare A_J absolute. It means only: for this receipt, these premises constitute the declared boundary of adjudication. Any admitted item can later be reopened as a new Claim Kernel. The stop rule is local and does not, by itself, guarantee finite termination of unrestricted recursive decomposition; a global termination claim additionally requires an effective stopping condition to be reached or another declared well-founded descent criterion.

11. Scoped Verification Receipts

A generalized receipt is

$$\mathfrak{R} = (\text{id}, \mathcal{K}, J, A_{\text{rec}}, K_{\text{rec}}^{\checkmark}, U_{\text{unc}}, \rho_{\text{rec}}, \Sigma), \tag{54}$$

where A_{rec} records the generalized receipt's admitted assumptions, $K_{\text{rec}}^{\checkmark}$ its verified content, U_{unc} records uncertainty or conditionality, ρ_{rec} records unresolved receipt residue, and Σ records dependencies on other receipts.

The formal receipt of Definition 7.1 is the formal-jurisdiction projection of this generalized receipt, not a second incompatible receipt type. When the generalized receipt concerns a formal jurisdiction J_F, identify $\mathscr{K} \mapsto \mathscr{K}$, $J \mapsto J_F$, $A_{\mathrm{rec}} \mapsto A_F$, $K^{\checkmark}_{\mathrm{rec}} \mapsto K^{\checkmark}_F$, and $\rho_{\mathrm{rec}} \mapsto \rho_F$. The generalized fields id, U_{unc}, and Σ remain available in the full receipt even when the reduced formal view $\mathfrak{R}_F$ suppresses them. Thus $\mathfrak{R}_F$ is a typed projection of $\mathfrak{R}$, not an unrelated tuple.

A receipt should answer:

1. What exact claim was checked?
2. Under what definitions and premises?
3. By which adjudicator?
4. What passed?
5. What failed?
6. What remains unresolved?
7. Which bridges were outside jurisdiction?
8. Which other receipts are dependencies?

Rule 11.1 — Receipt composition

Receipts $\mathfrak{R}_1$ and $\mathfrak{R}_2$ are declared composable only where the verified output type, assumptions, uncertainty conditions, and translation requirements of $\mathfrak{R}_1$ satisfy the declared input contract of $\mathfrak{R}_2$. This is a protocol rule, not a theorem.

This prevents a proof receipt from automatically composing with an ontological conclusion or an empirical receipt from automatically composing with a unique causal interpretation.

12. Worked Case I: Baryon Asymmetry as Problem Formation

The baryon example is useful only if this paper applies its own measurement-registry discipline. The quantity ordinarily quoted as an observational baryon-to-photon ratio is not an unmediated sensory datum. It is an inferred parameter produced from astronomical abundance observations through a physical and statistical registration chain.

Let

$$X_{\mathrm{BBN}} = \{\text{light-element abundance observations and associated measurement records}\}. \tag{55}$$

Let $\mathcal{M}_{\mathrm{BBN}}$ denote the declared registration map containing the abundance likelihood, nuclear/thermal model, photon-temperature convention, calibration inputs, and parameter-extraction procedure. Then the scoped observational anchor is

$$\hat{\eta}_B = q_{\mathcal{M}_{\mathrm{BBN}}}(X_{\mathrm{BBN}}). \tag{56}$$

The companion paper uses the 2024 Particle Data Group astrophysical-constants table value

$$\hat{\eta}_B = (6.04 \pm 0.12) \times 10^{-10}, \tag{57}$$

explicitly as a BBN-inferred, model-mediated anchor. The present paper therefore treats this value as a **registered empirical estimate**, not a bare datum.

The net baryon density is exactly

$$n_B = n_b - n_{\bar{b}}, \tag{58}$$

and the baryon-to-photon ratio is

$$\eta_B = \frac{n_B}{n_\gamma}. \tag{59}$$

After baryon-antibaryon annihilation, $n_{\bar{b}} \ll n_b$, so $\eta_B \simeq n_b/n_\gamma$. Here "after baryon-antibaryon annihilation" refers specifically to the approximation $n_{\bar{b}} \ll n_b$; it should not be conflated with electron-positron annihilation. Photon heating and the convention for comparing η_B across those epochs belong to the registration map $\mathcal{M}_{\mathrm{BBN}}$, not to the bare algebraic definition of the baryon-to-photon ratio.

12.1 Conventional problem kernel

A conventional baryogenesis traversal can be schematized as

$$\hat{\eta}_B + \mathcal{S}_{\mathrm{sym}} + \mathcal{O}_{\mathrm{dyn}} \mapsto P_{\mathrm{bar}}, \tag{60}$$

where $\mathcal{S}_{\mathrm{sym}}$ supplies the relevant symmetry/background structure and $\mathcal{O}_{\mathrm{dyn}}$, an explicitly declared dynamical specialization of the general ordering component $\mathcal{O}_{\mathrm{ord}}$, supplies a dynamical explanatory ordering. Sakharov's conditions organize the resulting mechanism-seeking problem.

Nothing in the present method disputes that program. The audit asks which content belongs to the registered empirical estimate, which belongs to $\mathcal{M}_{\mathrm{BBN}}$, and which enters only when the estimate is converted into the particular explanatory problem P_{bar}.

12.2 Lambda registry traversal

The complementary Lambda traversal holds the registered empirical target fixed while declining to grant every symmetry/reference choice primitive explanatory priority. It asks which portions of the stated problem survive translation between declared registries.

Let

$$\tau_{\mathrm{std}\to\Lambda} : \mathcal{G}_{\mathrm{std}} \dashrightarrow \mathcal{G}_{\Lambda} \tag{61}$$

be a partial translation map. A quantity q is invariant on the translated domain when

$$q_{\Lambda}(\tau_{\mathrm{std}\to\Lambda}(x)) = q_{\mathrm{std}}(x). \tag{62}$$

The directional translation residue is

$$\mathrm{Res}_{\mathrm{std}\to\Lambda}(\tau_{\mathrm{std}\to\Lambda}) = \mathfrak{S}_{\Lambda} \setminus \mathrm{Im}\left(\tau_{\mathrm{std}\to\Lambda}\right), \tag{63}$$

where, for this instance of the general typed translation-residue definition of Section 26, the declared target structure is $S_B = \mathfrak{S}_{\Lambda}$.

The methodological result is not that conventional baryogenesis is false. It is:

$$\boxed{\text{solution to } P_{\mathcal{G}} \neq \text{proof that } \mathcal{G} \text{ uniquely generates the admissible problem}}. \tag{64}$$

The companion paper is therefore a worked example of the upstream audit rather than a premise required for accepting the present method.

13. Worked Case II: Quantum Vacuum, Zero-Point Language, and the Casimir Interpretation Boundary

Quantum theory supplies a downstream example because several exact meanings of *zero* are often compressed in ordinary language.

13.1 Single-mode oscillator: one degree of freedom

For a quantum harmonic oscillator,

$$H|n\rangle = \hbar\omega\left(n + \frac{1}{2}\right)|n\rangle, \tag{65}$$

so the ground state obeys

$$E_0 = \frac{1}{2}\hbar\omega. \tag{66}$$

The state label $|0\rangle$, the excitation number $n = 0$, and a zero-valued physical observable are different typed statements.

For an oscillator coordinate x,

$$\langle 0|x|0\rangle = 0, \tag{67}$$

while

$$\langle 0|x^2|0\rangle > 0, \qquad \mathrm{Var}_0(x) > 0. \tag{68}$$

The preceding expectation-value and variance relations license exact statements about the oscillator state. They do not by themselves license an ontological sentence such as "nothing fluctuates into something."

13.2 Field theory requires an additional bridge

A quantum field $\phi(x)$ is not merely the oscillator coordinate x with a new name. Local products such as $\phi(x)^2$ generally require regularization/renormalization; the coincident two-point function may be ultraviolet singular. The audit therefore marks the transition

$$\text{single oscillator} \longrightarrow \text{quantum field} \tag{69}$$

as an explicit formal bridge rather than silently reusing the oscillator variance statement.

Well-defined field-theoretic statements are instead made with appropriately smeared fields, regulated correlation functions, normal ordering where appropriate, renormalized composite operators, or finite differences between configurations. The correct formal object depends on the theory and measurement problem.

13.3 Constant energy shifts

For ordinary non-gravitational quantum mechanics, let

$$H' = H + c_0 I. \tag{70}$$

Then

$$E'_n = E_n + c_0, \qquad E'_m - E'_n = E_m - E_n. \tag{71}$$

The absolute offset and observable transition differences therefore occupy different roles. When gravity or another coupling makes an absolute energy density operationally relevant, that additional coupling is itself a bridge requiring explicit treatment.

13.4 Vacuum language as a claim graph

A common linguistic chain can resemble

$$\text{ground state} \to \text{vacuum} \to \text{zero} \to \text{empty} \to \text{nothing} \to \text{fluctuation of nothing}. \tag{72}$$

This linguistic chain is not offered as a description of expert quantum field theory. It is an audit target for places where distinct formal, physical, and philosophical predicates are allowed to inherit one another's authority.

13.5 External case: Jaffe on the Casimir effect

An external published dispute supplies a useful stress test. R. L. Jaffe argued that Casimir forces can be formulated without treating zero-point energies as directly observable physical substances. In his parallel-plate analysis the force vanishes as the electromagnetic coupling $\alpha \to 0$, while the familiar ideal-conductor result is recovered in the strong-coupling limit $\alpha \to \infty$.

The methodological value of this example is not that Jaffe's interpretation must be adopted. It is that the same measured Casimir force can be embedded in different explanatory derivations. Therefore the empirical fact of a Casimir force should not automatically be presented as a receipt for every ontological claim attached to "vacuum energy."

The claim-routing system separates at least:

$$\text{measured force} \to \text{QED/scattering derivation} \to \text{zero-point representation} \to \text{ontological interpretation.} \tag{73}$$

Each arrow may be valid, useful, or conventional; none receives certification merely by proximity to the others.

14. Non-Promotion Stress Tests: Model Extensions, Object Attribution, and Exotic Physical Language

The preceding vacuum example motivates a broader audit principle. A successful formal representation, fitted extension, tracked feature, or technically legitimate noun does not automatically acquire a stronger causal or ontological type. The following cases test that principle without adjudicating the underlying interpretations in advance.

14.1 Closure degrees of freedom and the "hidden variable" compression

Suppose a restricted model leaves a registered residue

$$\mathrm{Res}_n^{\mathrm{mod}}(x; D) \neq 0. \tag{74}$$

An enlarged accounting may introduce an additional coordinate ξ and obtain

$$\mathrm{Res}_{n+1}^{\mathrm{mod}}(x, \xi^*; D) = 0. \tag{75}$$

What follows immediately is a statement about the enlarged description: it admits closure unavailable to the restricted description. It does not follow merely from that closure that ξ^* has been independently registered as a physical entity, that it uniquely causes the former residue, or that no other extension could close the same accounting.

The conventional phrase "hidden variable" can compress several different claims: a coordinate absent from a chosen representation, a quantity unresolved by an instrument, a latent model parameter, a variable inaccessible under a projection, or an asserted unobserved physical cause. The audit does not select one meaning by default. It decomposes the phrase and asks which status has actually been warranted.

The quantum-mechanical use of "hidden variable" carries additional formal constraints and therefore must not be treated as an unconstrained placeholder. Bell's theorem does not rule out every possible hidden-variable formulation; it constrains theories satisfying the locality/factorization assumptions used to derive Bell inequalities, because those theories cannot reproduce all quantum correlations. Kochen–Specker independently constrains noncontextual assignments of definite values that preserve the functional relations required by quantum observables. Later loophole-closing Bell tests supplied discriminating empirical consequences not used merely to construct a model closure. These results are therefore examples of the present rule rather than exceptions to it: a representational extension gains empirical constraint when competing structures entail measurably different statistics.

Accordingly, let X_n denote their common lower-resolution description. Let two extended state spaces $\widetilde{X}_1$ and $\widetilde{X}_2$ contain states $\tilde{x}_1 \in \widetilde{X}_1$ and $\tilde{x}_2 \in \widetilde{X}_2$, with declared projections $\pi_1 : \widetilde{X}_1 \to X_n$ and $\pi_2 : \widetilde{X}_2 \to X_n$. To compare a discriminating observable across the two extended spaces without identifying those spaces by notation alone, declare a common codomain Y_r and an observable $r : \widetilde{X}_1 \sqcup \widetilde{X}_2 \to Y_r$. A discriminating surplus relative to the common lower-resolution description requires both lower-description coincidence and a separately registered distinction in that common codomain:

$$\tilde{x}_1 \in \widetilde{X}_1, \qquad \tilde{x}_2 \in \widetilde{X}_2, \qquad \pi_1(\tilde{x}_1) = \pi_2(\tilde{x}_2), \qquad r(\tilde{x}_1) \neq r(\tilde{x}_2). \tag{76}$$

Then the difference between the extensions is no longer merely surplus accounting relative to the lower projection. It has acquired a discriminating test surface. What the test excludes remains scoped to the assumptions of the theorem and experiment; it does not license the broader sentence "all hidden variables are impossible."

A lift makes the representational point explicit. Let

$$L : X_n \to X_{n+1}, \qquad \pi : X_{n+1} \to X_n. \tag{77}$$

Distinct lifted states can satisfy

$$\tilde{x}_1 \neq \tilde{x}_2, \qquad \pi(\tilde{x}_1) = \pi(\tilde{x}_2). \tag{78}$$

The lower-dimensional accounting then lacks a distinction available upstairs. Introducing the additional degree of freedom provides a new way to zero the residual; it does not by itself identify the ontology of that degree of freedom. A stronger case arises only when the extension yields discriminating consequences not used to construct the closure.

14.2 The laser spot, apparent motion, and UAP-style object attribution

A laser spot chased by a cat supplies a deliberately ordinary example. The luminous locus can be tracked as a position $x_{\text{spot}}(t)$, and its apparent velocity and acceleration can be calculated:

$$v_{\text{spot}} = \dot{x}_{\text{spot}}, \qquad a_{\text{spot}} = \ddot{x}_{\text{spot}}. \tag{79}$$

Those kinematic quantities can correctly describe the motion of the registered locus without establishing that the locus is a persistent massive object carrying a trajectory through space in the same sense as a thrown body. Adding a mass and applying a massive-body force interpretation requires another bridge.

The same typing discipline applies to disputed or unusual aerial observations without deciding what any particular observation is. A generic chain may run

$$\text{sensor record} \to \text{tracked feature} \to \text{persistent object} \to \text{massive object} \to \text{dynamics} \to \text{propulsion inference}. \tag{80}$$

An apparent trajectory may be accurately extracted while one or more later attributions remain open. Conversely, showing that an object attribution has not been established does not establish a preferred alternative such as projection, plasma, artifact, or field phenomenon. Each alternative carries its own bridge burden. The example demonstrates why objecthood and causal mechanism must not be smuggled into the data type by the vocabulary used to describe a track.

14.3 Exotic energy, exotic material, and nominalized explanations

Terms such as "zero-point energy," "vacuum energy," "negative energy," "exotic energy," and "exotic material" can be technically legitimate within a declared theory while also inviting semantic promotion when exported into broader claims. The audit therefore expands the noun into its prerequisites: which operator or stress-energy quantity is meant, relative to which reference or state, under which regularization or boundary conditions, in which geometry, and through which measurement or correspondence map does the quantity become operationally relevant?

This does not dismiss unconventional proposals. It prevents either enthusiasm or skepticism from substituting for the bridge. A mathematically coherent model of an unusual energy condition can remain coherent as a model even when its physical realization is unresolved; a registered effect can remain supported even when the proposed causal explanation is not uniquely identified.

15. Bidirectional Verification: Decompose Downward, Construct Upward

The preceding sections describe the downward audit of a claim. Existing constructive work supplies the complementary direction. The combined architecture is

$$\boxed{\text{claim} \xrightarrow{\mathfrak{D}} \text{declared primitive seam} \xrightarrow{\Gamma} \text{certified construction} \xrightarrow{B_{\text{out}}} \text{claimed meaning}}. \tag{81}$$

The primitive seam is intentionally local. It identifies the primitives accepted for the current adjudication; it does not assert that those primitives are ontologically fundamental.

15.1 Constructive traversal

Definition 15.1 — Finite constructive trace. Given a declared primitive set Π_{prim}, grammar Γ, and verifier Ver, a constructive traversal is a finite trace

$$C^{(0)} = \Pi_{\text{prim}}, \qquad C^{(k+1)} = \gamma_k(C^{(k)}), \quad \gamma_k \in \Gamma, \tag{82}$$

together with sufficient provenance to allow Ver to check every admitted step or a certificate reducible to those steps.

The output is not merely a conclusion. It is an addressable construction package containing the question, frame, inputs, operations, dependencies, intermediate objects, validation results, completion state, unresolved residue, and provenance.

15.2 Dependency and independence testing

Downward claim decomposition produces a declared premise set

$$\mathsf{Prem}(C) = \{p_1, \dots, p_n\}. \tag{83}$$

Constructive execution does not identify those premises with grammar primitives by notation alone. A declared compilation or seam map

$$\beta_{\text{pc}} : \mathsf{Prem}(C) \dashrightarrow \Pi_{\text{prim}} \tag{84}$$

records how admitted premises are represented, expanded, compiled, or discharged into the primitive basis used by the constructive grammar. The map may be partial: some premises can require imported certificates or external adjudication rather than direct primitive expansion.

An independence test is therefore performed on the constructive dependency actually used. Suppose a primitive dependency $u_j \in \Pi_{\text{prim}}$ is advertised as indispensable. The test is admissible only if the remaining grammar, macros, imported certificates, verifier obligations, and seam map are themselves u_j-free. Here Macro denotes the admitted macro library and Cert the imported-certificate dependency set:

$$u_j \notin \mathrm{Dep}(\Gamma, \mathsf{Ver}, \mathsf{Macro}, \mathsf{Cert}, \beta_{\text{pc}}). \tag{85}$$

If, under that condition,

$$\Pi_{\text{prim}} \setminus \{u_j\} \xrightarrow{\Gamma} Q, \tag{86}$$

then the successful reconstruction is **evidence** that the constructive description over-attributed necessity to u_j. It does not by itself prove that every source-level premise associated with u_j is philosophically or empirically dispensable.

Conversely, if removal of a declared dependency prevents reconstruction while all other admitted dependencies remain fixed, that failure is evidence that the dependency belongs to the scoped constructive burden. Both directions remain evidential outside the declared constructive class.

15.3 Multiple constructive paths and invariant results

Let two declared registries support certified outputs Q_A and Q_B:

$$\Pi_A \xrightarrow{\Gamma_A} Q_A, \qquad \Pi_B \xrightarrow{\Gamma_B} Q_B. \tag{87}$$

Do not identify Q_A and Q_B by notation alone. Let

$$\tau_{AB} : Q_A \dashrightarrow Q_B \tag{88}$$

be a declared partial translation. For a comparison observable q, constructive invariance means

$$q_B(\tau_{AB}(Q_A)) = q_A(Q_A) \tag{89}$$

where the translation is defined. Only under such an identity criterion may the two constructive paths be said to recover the same scoped invariant. The result does not prove that the registries are equivalent.

That distinction is central to complementary modeling.

15.4 IRRPROOF as a constructive implementation

The IRRPROOF architecture already separates four layers: claim, construction, validation, and residue. Its intended object model extends beyond theorem proofs to simulations, laboratory procedures, digital twins, circuits, protocols, and agent-role changes. This matters because the verifier appropriate to a claim depends on the claim's type. A theorem kernel, circuit simulator, calibration certificate, reproduction protocol, and plural jury do not possess interchangeable jurisdictions.

The present paper therefore does not treat IRRPROOF as merely another downstream theorem prover. It treats it as a constructive substrate capable of receiving a formalized Claim Kernel and returning a traceable object whose status can be recomposed into the larger claim graph.

16. Proof, Recognition, Reconstruction, and Trust

The word *proof* often conflates several different states.

16.1 Proof states and their relations

Let Obj be a submitted proof-bearing object, Ver a verifier, and H a human or independent machine auditor. Define:

$$\begin{aligned}
&\mathsf{PS}_1 : \mathsf{Ver}(\mathsf{Obj}) = \text{accept},\\
&\mathsf{PS}_2 : \mathsf{Obj} \text{ has a valid expansion into the verifier's admitted primitive/macro basis},\\
&\mathsf{PS}_3 : H \text{ can independently replay or inspect the entire required construction to a declared boundary},\\
&\mathsf{PS}_4 : H \text{ accepts the interpretation attached to the result.}
\end{aligned} \tag{90}$$

These relations are not equivalent.

When the auditor's declared reconstruction boundary refines, or is accompanied by a declared map sufficient to recover, the verifier's admitted primitive/macro basis, the scoped implication holds:

$$\mathsf{PS}_3 \Rightarrow \mathsf{PS}_2. \tag{91}$$

Without that boundary-compatibility condition, reconstructive accessibility relative to an auditor's chosen boundary does not by itself establish expansion into the verifier's admitted basis.

If Ver is complete for valid expansions over the declared admitted primitive/macro basis, then

$$\mathsf{PS}_2 \Rightarrow \mathsf{PS}_1. \tag{92}$$

If Ver is sound for that same admitted basis, acceptance licenses the converse scoped implication,

$$\mathsf{PS}_1 \Rightarrow \mathsf{PS}_2. \tag{93}$$

Hence a verifier that is both sound and complete on the declared class supports $\mathsf{PS}_1 \iff \mathsf{PS}_2$ on that class. Neither property is inferred merely from the existence of a verifier; each belongs to the verifier's own declared or independently established contract.

The remaining distinctions still do not collapse. A certificate that is valid under a sound and complete verifier may remain inaccessible to a particular reader, so PS_2 need not imply PS_3; a reader may trust an accepted certificate without independently reconstructing it; and interpretive assent PS_4 remains logically independent of the formal implication chain unless additional bridge premises are included.

16.2 Certificates and reconstruction

Definition 16.1 — Certificate validity.

Certificate validity is the scoped claim that a declared verifier accepts a proof or construction object under a declared kernel, rule set, and dependency stack.

Definition 16.2 — Reconstructive accessibility.

A certificate is reconstructively accessible relative to auditor H, primitive boundary Π_{prim}, and grammar Γ when H can independently replay or inspect the entire required construction to that declared boundary. When reconstructive accessibility is used to infer PS_2, that boundary must additionally refine, or carry a declared map sufficient to recover, the admitted primitive/macro basis of Ver; accessibility to an unrelated boundary carries no such implication:

$$\mathrm{RA}(\mathsf{Obj}; H, \Pi_{\text{prim}}, \Gamma) \in \{0, 1\}. \tag{94}$$

Version 1 intentionally keeps RA binary. Any graded reconstruction score would require a separately declared granularity or cost model, because raw node counts can be altered by macro packaging.

16.3 Oracle opacity

Define the oracle relation

$$\mathscr{O}_{\text{oracle}}(\mathsf{Obj}) \in \{\text{accept}, \text{reject}\} \tag{95}$$

when the consumer receives a verdict without a reconstructible trace. Such a verdict may be rationally trusted, but it is epistemically different from personally certifying the proof.

The architecture therefore prefers

$$\text{verdict} + \text{certificate} + \text{dependencies} + \text{primitive expansion} \tag{96}$$

over a bare badge.

17. Prior Constructive Instantiations within the Research Corpus

The bidirectional proposal did not originate solely from the present semantic analysis. Several prior instruments in the same research corpus already instantiate parts of its formal side. They are not independent external replications of this paper. Their value here is architectural continuity, and that non-independence is retained as explicit provenance rather than presented as validation.

17.1 The Relational Crucible

The Relational Crucible separates theorem-certified reachability from an agent's measured search behavior. When it reports a target unreachable, the negative result is not inferred from the agent's failure to find the target; an algebraic obstruction certificate exists independently of the run. The architecture therefore keeps at least four statuses separate: classical reachability facts, machine arithmetic, measured search behavior, and any interpretation of what that behavior means.

This is precisely the discipline proposed here. Search behavior is evidence about an agent. It is not silently promoted into a theorem about the registry.

17.2 Finite witness to infinite registry

The finite-witness protocol distinguishes finite computation from theorem-governed continuation. A finite browser does not enumerate an infinite object. It can calculate a finite witness, establish compatibility with a standard inverse system or theorem, and then render the continuation as theorem-governed rather than as computationally exhausted.

This gives a concrete model for handling limits:

$$\text{finite witness} \neq \text{infinite completion}, \tag{97}$$

while still allowing the finite witness to support a rigorous statement about the infinite registry when an explicit theorem supplies the bridge.

The same discipline applies to measurement nulls and limiting zeros: a finite station can constrain a limit without becoming the limit.

17.3 Positional recoverability

The positional-recoverability work asks which structural relations can be recovered from an addressing registry alone and which require supplementary information. Its symmetry obstruction shows that a symmetric addressing vocabulary cannot recover an antisymmetric relation without additional data. The worked Cartan family then reaches the resulting lower bound across all finite types.

Methodologically, define the recoverable portion of a target structure Tgt under grammar Γ as $\mathrm{Rec}_\Gamma(\mathsf{Tgt})$. The supplementary residue is

$$\mathrm{Supp}_\Gamma(\mathsf{Tgt}) := \mathsf{Tgt} \setminus \mathrm{Rec}_\Gamma(\mathsf{Tgt}). \tag{98}$$

When $\mathrm{Rec}_\Gamma(\mathsf{Tgt}) \subseteq \mathsf{Tgt}$, the definition above yields the disjoint decomposition

$$\mathsf{Tgt} = \mathrm{Rec}_\Gamma(\mathsf{Tgt}) \sqcup \mathrm{Supp}_\Gamma(\mathsf{Tgt}). \tag{99}$$

The supplement is not treated as a defect of the target. It measures what the selected representation fails to carry positionally.

17.4 DST and QED completion

DST supplies plural review under declared canons, premises, stopping conditions, and unresolved residue. QED completion asks whether a segment satisfies its declared completion contract rather than whether it has attained frame-independent final truth.

Together these systems support the present routing principle:

$$\text{different claim type} \Rightarrow \text{different appropriate adjudication}. \tag{100}$$

18. The Zero/Null Case as a Full-Stack Demonstration

The zero audit now acquires a constructive counterpart.

Consider the statement:

> The experiment observed zero.

Downward decomposition asks:

1. What observable was coupled to the detector?
2. What baseline x_0 was selected?
3. What resolution $\epsilon_{\mathcal{M}}$ was available?
4. What does the zero denote: algebraic identity, limiting value, state/excitation label, measured estimate, ontological absence claim, or another declared denotation?
5. What role does that zero perform: coordinate origin, reference level, calibration baseline, renormalization convention, symmetry/equilibrium ideal, experimental null threshold, or another declared role?
6. What linguistic bridge converted the instrument record into the word *zero*?

The formalized empirical kernel may reduce to

$$|\hat{x} - x_0| < \epsilon_{\mathcal{M}}. \tag{101}$$

That kernel can be mechanically checked against instrument logs, calibration data, uncertainty propagation, and the declared statistical rule.

The interpretive statement

$$x = 0_{\text{ont}} \tag{102}$$

does not follow unless an additional bridge is supplied.

The same structure appears in computation. A numerical routine can return a machine zero because a quantity underflows, rounds, cancels within finite precision, or equals the algebraic identity exactly. Those events have different certificates.

Likewise, a limit statement

$$\lim_{n\to\infty} a_n = 0 \tag{103}$$

can be certified from a finite proof object without any finite a_n being equal to zero.

Thus the zero case unifies measurement, enumeration, computation, limits, and semantics. The token is stable; the type is not.

19. Companion Case: Lambda as Complementary Modeling

The companion paper *Lambda as Complementary Modeling: Registry-Relative Symmetry, Joint Closure, and the Baryon-Photon Ratio* can be read as an application of the present method.

Its central move is to hold the registered empirical estimate fixed while separating:

$$D = \hat{\eta}_B \tag{104}$$

from the problem-forming commitments used to ask what that datum requires.

The conventional baryogenesis traversal, the TuFT traversal, and the Lambda registry traversal can then be represented as different declared paths toward a shared target. The purpose is not to award one path exclusive ontological status. It is to determine which quantities remain invariant, which assumptions belong to each registry, which bridges are required for interpretation, and which residue appears under translation.

Accordingly, the companion paper is not merely another example cited by this framework. It is a worked case of the upstream audit:

$$\begin{aligned}\text{shared registered estimate} &\to \text{different problem-forming registries}\\ &\to \text{scoped formal/physical traversals}\\ &\to \text{explicit interpretive comparison.}\end{aligned} \tag{105}$$

The present paper generalizes that discipline beyond baryon asymmetry.

20. Canon, Convention, and Robustness

Bodies of accepted knowledge contain more than proved propositions. They also contain measurement conventions, representational choices, terminology, standard approximations, model correspondences, and methodological expectations.

A convention may evolve through

$$\text{selection} \to \text{formalization} \to \text{application} \to \text{stress} \to \text{revision} \to \text{stabilization.} \tag{106}$$

The mature result may be extremely robust. The method therefore distinguishes two propositions:

$$K_{\mathrm{rob}}: \quad \text{the convention is robust and productive in its domain,} \tag{107}$$

and

$$K_{\mathrm{uniq}}: \quad \text{the convention is the uniquely necessary representation of the target.} \tag{108}$$

Evidence for K_{rob} does not automatically prove K_{uniq}.

This distinction applies symmetrically to canonical and noncanonical research. An alternative proposal does not receive credit merely for being unconventional. It must expose its primitives and survive the same routing. Conversely, a canonical claim does not bypass decomposition merely because its vocabulary is familiar.

20.1 Identity, excluded middle, and alternative accounting roles

The framework does not treat one identity convention as the universal competitor of all others. An identity relation is useful because it preserves selected sameness while permitting other differences to be ignored for a task. Classical excluded middle likewise remains valid and powerful inside a logic that declares it. The audit question is not whether bivalence should be abolished, but whether exhaustiveness inside a declared predicate space is being exported as exhaustiveness of every admissible source description.

The same caution applies to terms such as “transcendental.” In a formal context, transcendence is defined relative to a specified algebraic class; it should not be silently promoted into a claim of metaphysical ineffability. A boundary of one identity or representation system may be an ordinary object of another.

The project’s prior Euler/Germain comparison supplies an instructive algebraic example. Different exact factorizations can preserve different structural relations and support different accounting tasks without requiring one representation to invalidate the other. The methodological point used here is only this: when two exact descriptions organize identity differently, assessment should ask which invariants, transitions, and tasks each representation preserves rather than assuming that notational singularity is itself the target.

20.2 Dependency cycles are typed, not automatically disqualifying

The word “circular” also compresses distinct structures. Recursive definitions, fixed points, mutually constrained systems, axiomatic closure, and justificatory dependence are not the same defect. For example,

$$x_{k+1} = g(x_k), \qquad x^* = g(x^*) \tag{109}$$

are ordinary recursive and fixed-point forms, where g is a local map in this example and is not the registered traversal F. Likewise a theorem statement

$$\mathsf{Prem} \vdash_{\Gamma} C \tag{110}$$

records derivability of the scoped claim C from a declared premise set under the registered grammar Γ; it does not purport to establish those commitments from nowhere.

A dependency cycle becomes epistemically defective for this method when a result is presented as warrant independent of premises or dependencies on which that result itself relies, unless the declared warrant explicitly permits that dependence. The audit question is therefore not merely "is there a loop?" but "what is this loop being asked to establish?"

21. Claim Graphs and Automated Routing

The method is designed to become computational infrastructure.

Let a document W be transformed into a directed claim graph

$$G_W = (V_G, E_G), \tag{111}$$

where each node $v_i \in V_G$ is a Claim Kernel and each edge $e_{ij} \in E_G$ records a dependency, inference, correspondence, or interpretive bridge.

Let $\mathcal{J}$ be the set of available adjudicator classes. Routing is partial and set-valued:

$$\mathscr{R}_{\text{route}} : \mathcal{T} \rightharpoonup \text{Set}(\mathcal{J}). \tag{112}$$

A well-typed kernel may route to several compatible adjudicators; an unresolved mixed kernel may be left undefined and returned for decomposition rather than forced into a single route.

A practical routing table is:

Kernel type	Primary routing
Formal F	theorem prover, symbolic checker, proof audit, DST / Theory-of-Everything (TOE) formal traversal
Registration / empirical-report E	registration provenance, statistics, replication, experimental comparison
Semantic S	definition comparison, ambiguity/equivocation audit
Methodological M	protocol and convention comparison
Bridge B	correspondence and assumption audit
Reference/zero/limit Z	invariance, calibration, limit, and zero-type audit

The system should not force a claim into one adjudicator when its type remains mixed. It should decompose again.

21.1 Neutrality by construction

Authority can be relevant evidence about reliability, provenance, or prior scrutiny. It is not a substitute for the formal content of a claim. A neutral arena can therefore preserve source metadata while ensuring that the formal adjudicator receives the same kernel schema for canonical and independent work.

Here *neutrality* means neutrality of the routing and adjudication protocol with respect to source status once a kernel has been typed; it does not mean that every example used to illustrate the architecture is an evidentially independent validation of it. IRRPROOF, DST, QED completion, and TOE Arena are in-corpus implementations or applications and are treated as architectural instantiations, not as independent replication.

This is particularly useful for independent research communities. The method offers neither automatic validation nor automatic dismissal. It offers a common interface.

22. Integration with Existing Adjudication Systems

The routing layer is implementation-independent, but it can feed existing or developing infrastructures.

A formalized kernel can be translated into an IRRPROOF construction object whose claim, construction, validation, and residue layers remain separately addressable. Depending on type, that object can then route into:

- a DST traversal with explicit premises, gates, flags, and frontier conditions;
- a TOE Arena entry in which competing well-formed arguments are compared under the same rules;
- a QED-style bounty submission with a declared target and falsification surface;
- a Lean, Coq, Isabelle, or other proof-assistant statement when the claim admits formal encoding;
- numerical reproduction or symbolic-algebra checks;
- empirical receipts tied to datasets and uncertainty models.

The important architectural rule is that downstream systems return scoped receipts to the claim graph rather than a single global verdict.

A document can therefore contain simultaneously:

$$\text{proved}, \quad \text{empirically supported}, \quad \text{interpretive}, \quad \text{open}, \quad \text{failed}, \tag{113}$$

without contradiction.

23. Failure Modes and Falsification Criteria for the Method

A method that audits other arguments should expose its own failure conditions.

23.1 Non-unique decomposition

Two competent auditors may produce materially different but individually adequate kernelizations:

$$G_C^{(1)} \neq G_C^{(2)}, \qquad \mathrm{Adeq}(G_C^{(1)}; C) = \mathrm{Adeq}(G_C^{(2)}; C) = 1. \tag{114}$$

This is not automatically a failure. It is a measurable decomposition ambiguity. If the disagreement cannot itself be represented as explicit competing kernels, or if auditors cannot agree on the adequacy criteria being applied, the procedure has failed to provide the promised clarity.

23.2 Bridge proliferation

Recursive decomposition can generate more bridges than it resolves. The Declared Primitive Stop Rule controls local termination, but excessive proliferation may make the method impractical. Complexity metrics should therefore be reported.

23.3 False precision

Typing a philosophical statement does not solve it. A semantic label must not masquerade as adjudication.

23.4 Misclassification

Automated routing can classify a formal claim as interpretive or an interpretive bridge as formal. Human review and reversible routing are therefore required for high-stakes use.

23.5 Receipt laundering

A receipt can itself be misused. Systems must prevent a narrow receipt from being presented as global validation. Machine-readable scope and dependencies should accompany every badge or stamp.

23.6 Empirical evaluation

The method makes testable process claims. Its primary empirical endpoint is the **unflagged invalid cross-type inference rate** on a preregistered benchmark corpus whose candidate cross-type transitions have independently adjudicated validity labels. To prevent a procedure from improving this endpoint merely by flagging every transition, the primary endpoint is reported jointly with a **false-flag rate** over adjudicated valid transitions. Define

$$E_{\text{cross}} = \frac{N_{\text{invalid, unflagged}}}{N_{\text{invalid, adjudicated}}}, \qquad F_{\text{flag}} = \frac{N_{\text{valid, flagged}}}{N_{\text{valid, adjudicated}}}. \tag{115}$$

A benchmark is eligible for the paired primary test only when both $N_{\text{invalid, adjudicated}} > 0$ and $N_{\text{valid, adjudicated}} > 0$. A control set with either denominator equal to zero may still be reported diagnostically, but the corresponding rate is recorded as not applicable rather than assigned an arbitrary numerical value.

Before evaluation, the benchmark, baseline procedure, improvement margin $\delta_{\text{cross}} > 0$, absolute maximum admissible false-flag rate $\varepsilon_{\text{flag}} \in [0,1]$, maximum baseline-relative false-flag degradation $\delta_{\text{flag}} \geq 0$, and uncertainty criterion must be declared. The improvement endpoint is eligible exactly when the benchmark itself is eligible and $E^{\text{baseline}}_{\text{cross}} \geq \delta_{\text{cross}}$. If the benchmark is eligible but $E^{\text{baseline}}_{\text{cross}} < \delta_{\text{cross}}$, the *paired primary process claim as a whole* is reported as not applicable for that benchmark; F_{flag} may still be reported diagnostically but cannot alone produce a pass or failure of the paired claim. The equality case $E^{\text{baseline}}_{\text{cross}} = \delta_{\text{cross}}$ is intentionally eligible and demands zero unflagged invalid transitions.

When the paired primary process claim is eligible, it is

$$E^{\text{method}}_{\text{cross}} \leq E^{\text{baseline}}_{\text{cross}} - \delta_{\text{cross}} \quad \wedge \quad F^{\text{method}}_{\text{flag}} \leq \min\left\{\varepsilon_{\text{flag}}, F^{\text{baseline}}_{\text{flag}} + \delta_{\text{flag}}\right\}. \tag{116}$$

For an eligible paired claim, failure of either preregistered inequality rejects that claim for the benchmark. Success supports only the scoped claim that the method reduced unflagged invalid cross-type inference without exceeding the declared false-flag tolerance; it does not by itself establish universality beyond the tested domain. The false-flag requirement intentionally has no separate improvement-margin gate beyond benchmark denominator eligibility and the preregistered absolute/baseline-relative tolerances.

Secondary diagnostics may include:

- inter-auditor agreement about what was actually proved;
- detection of unstated assumptions;
- reproducibility of formal subclaims;
- clarity about unresolved disagreement;
- successful routing to appropriate verification tools.

These secondary measures diagnose where the method helps or fails, but they do not substitute for the preregistered primary endpoint.

For inter-auditor comparison, raw node identity is not the primary metric because graph granularity can vary without changing reconstructive burden. Use instead a declared pairwise adequacy-comparison predicate

$$\mathrm{Cmp}_{\mathrm{Adeq},C}(G_1, G_2; \mu_{\text{cmp}}) = 1, \tag{117}$$

where μ_{cmp} is the comparison map declared for that reporting task. The predicate records that both graphs are adequate for the same scoped claim C and, under μ_{cmp}, preserve the same declared reconstructible dependencies, bridge obligations, exported claim types, and declared leaf-stopping assignments $\mathsf{LeafStop}_C$ (including jurisdiction, stop mode, and actor where required). No reflexivity, symmetry, or transitivity is presumed beyond what a particular comparison protocol separately establishes. Inter-auditor agreement is therefore reported pairwise under the declared μ_{cmp}, rather than over undeclared equivalence classes or raw node matches.

24. A Minimal Operational Protocol

The complete procedure can be stated compactly.

Step 1 — Register. Identify the source observation, stipulated input, or target relation and the measurement/representation registry.

Step 2 — Decompose. Extract atomic or near-atomic claims and construct Claim Kernels.

Step 3 — Audit problem formation. Identify the primitives, references, equivalences, symmetries, causal orderings, and calibration choices required to transform the datum into the stated problem.

Step 4 — Type. Assign provisional formal, empirical, semantic, methodological, bridge, and reference types.

Step 5 — Route. Send each kernel only to adjudicators with matching jurisdiction.

Step 6 — Receipt. Return verified content, assumptions, uncertainty, failures, and unresolved residue.

Step 7 — Audit interpretation. Identify the bridges connecting verified results to broader conclusions.

Step 8 — Recurse. Promote unresolved bridges or disputed primitives into new Claim Kernels.

Step 9 — Stop locally. Apply the Declared Primitive Stop Rule for the current receipt.

Step 10 — Recompose. Reconstruct the document-level claim graph without erasing the different statuses of its components.

The output is not a binary verdict. It is a typed map of what the argument establishes and what it additionally assumes.

25. Worked Self-Application: Proposition 3.2

The method should expose at least one of its own formal claims to the same object model it proposes for other work. Consider Proposition 3.2, the bridge-underdetermination lemma. To avoid encoding the proposition's conditionality both in the claim text and again in the Claim Kernel premise field, let C_{BU} denote only its scoped conclusion:

> Acceptance of the shared verifier-visible formal traversal underdetermines the declared adequacy property of the two upstream bridges.

The hypotheses that license this conclusion are carried explicitly by $\mathsf{Prem}_{\mathrm{BU}}$, as a Claim Kernel is designed to do. A filled Claim Kernel is

$$\mathcal{K}(C_{\mathrm{BU}}) = (C_{\mathrm{BU}}, F, \mathsf{Prem}_{\mathrm{BU}}, \mathsf{Def}_{\mathrm{BU}}, \Gamma_{\mathrm{BU}}, J_{\mathrm{BU}}, \mathcal{X}_{\mathrm{BU}}, \rho_{\mathrm{BU}})\,, \tag{118}$$

with the following instantiated fields:

- $\mathsf{Prem}_{\mathrm{BU}}$: Ch_1 and Ch_2 are well-formed registration chains of Definition 3.1; they share the same formal traversal F and the same verifier-visible tuple $(P, \mathsf{Prem}_F, \Gamma_F)$; the scoped verification relation depends only on that tuple and yields the same accepted result Q in both chains; and the declared bridge-adequacy assignments satisfy $A_{\mathrm{in}}(B_{\mathrm{in}}^{(1)}) = 1$ and $A_{\mathrm{in}}(B_{\mathrm{in}}^{(2)}) = 0$. This mirrors Proposition 3.2's stated hypotheses rather than replacing one with a derived or weaker modal condition. The common-Q clause is retained intentionally: although it is derivable from tuple identity together with verifier dependence on that tuple, $\mathsf{Prem}_{\mathrm{BU}}$ mirrors Proposition 3.2 as stated rather than claiming a logically minimal premise basis.
- $\mathsf{Def}_{\mathrm{BU}}$: registration chain, verifier input, upstream bridge, and bridge-adequacy predicate have the meanings fixed in Sections 3 and 7; in particular, A_{in} is a property of the upstream bridge and is not a component of the verifier-visible tuple $(P, \mathsf{Prem}_F, \Gamma_F)$.
- Γ_{BU}: equality/substitution, tuple comparison, verifier-input dependency reasoning, and the bridge-underdetermination inference used in the proof of Proposition 3.2.
- J_{BU}: formal/metalogical jurisdiction over what follows from the declared verifier input.
- $\mathcal{X}_{\mathrm{BU}}$: links to any substantive empirical, semantic, or methodological criteria used to decide whether a particular real-world bridge satisfies A_{in}.
- ρ_{BU}: the actual adequacy of any particular real-world bridge, uniqueness of an admissible bridge, and any broader ontological interpretation are not decided by the proposition.

The corresponding scoped receipt instantiates the same presentation fields declared in Appendix D:

> **RECEIPT ID:** R-BU-001
> **STATUS:** VERIFIED — FORMAL JURISDICTION ONLY
> **KERNEL:** $\mathcal{K}(C_{\mathrm{BU}})$
> **JURISDICTION:** Formal dependency/metalogical claim.
> **PREMISES:** $\mathsf{Prem}_{\mathrm{BU}}$, as instantiated above.
> **CHECKED:** Under the stated premises, verifier acceptance of the common formal traversal underdetermines an upstream bridge property omitted from the verifier input.
> **UNCERTAINTY / CONDITIONALITY:** Conditional only on the declared premises, definitions, and verifier-input boundary; no additional numerical uncertainty is asserted by this formal receipt.
> **NOT CHECKED:** Whether either concrete bridge is empirically adequate, semantically preferable, causally correct, or ontologically unique.
> **RESIDUE:** Any such bridge assessment remains a linked Claim Kernel with its own adjudication requirements.
> **DEPENDENCIES:** Proposition 3.2; Definitions 3.1 and 7.1; the declared meanings of A_{in}, $\mathsf{Prem}_{\mathrm{BU}}$, and Γ_{BU}.

This self-application demonstrates the intended non-promotion rule. The proposition can be formally valid without the paper pretending that the proposition adjudicates the external bridges it discusses. Conversely, leaving those bridge questions outside this receipt does not weaken the formal result. The kernel is suitable for proof-assistant encoding because its formal burden is finite and explicit; this manuscript does not claim that this particular kernel has already been certified by Lean.

25.1 Worked kernel decomposition of the same claim

The same claim also supplies a finite worked instance of Definition 5.2 rather than leaving adequacy only at the level of specification. Let

$$G_{\mathrm{BU}} = (V_{\mathrm{BU}}, E_{\mathrm{BU}}), \tag{119}$$

with

$$V_{\mathrm{BU}} = \{\mathcal{K}(C_{\mathrm{BU}}), \mathcal{K}(C_{\mathrm{vis}}), \mathcal{K}(C_{\mathrm{bridge}})\}, \tag{120}$$

where C_{vis} records the verifier-side premises: both chains are well formed, they share the same formal traversal F, the verifier-visible tuples $(P, \mathsf{Prem}_F, \Gamma_F)$ are equal, and the scoped verification relation depends only on that tuple and yields the same accepted result Q in both chains. The second leaf, C_{bridge}, records the definite bridge-side premise $A_{\mathrm{in}}(B^{(1)}_{\mathrm{in}}) = 1$ and $A_{\mathrm{in}}(B^{(2)}_{\mathrm{in}}) = 0$. Both leaves have formal type: $T(C_{\mathrm{vis}}) = F$ and $T(C_{\mathrm{bridge}}) = F$.

Orient the decomposition edges downward as required by Definition 5.2:

$$E_{\mathrm{BU}} = \{\mathcal{K}(C_{\mathrm{BU}}) \to \mathcal{K}(C_{\mathrm{vis}}),\ \mathcal{K}(C_{\mathrm{BU}}) \to \mathcal{K}(C_{\mathrm{bridge}})\}. \tag{121}$$

Thus the declared leaf boundary is

$$L_{\mathrm{BU}} = \{\mathcal{K}(C_{\mathrm{vis}}), \mathcal{K}(C_{\mathrm{bridge}})\}. \tag{122}$$

For this worked formal jurisdiction, declare $A_{J_{\mathrm{BU}}}$ independently as the syntactically defined set of premise forms admitted by the finite grammar of Γ_{BU}. Membership in $A_{J_{\mathrm{BU}}}$ is decidable by parsing a proposed premise form and checking it against that finite grammar; the boundary is fixed before either leaf is tested. The two leaf premise sets are finite and every premise form appearing in them passes this membership test. Hence the effective-stop preconditions of Definition 10.1 hold and

$$\mathrm{Stop}^{\mathrm{eff}}_{J_{\mathrm{BU}}}(\mathcal{K}(C_{\mathrm{vis}})) = 1, \qquad \mathrm{Stop}^{\mathrm{eff}}_{J_{\mathrm{BU}}}(\mathcal{K}(C_{\mathrm{bridge}})) = 1. \tag{123}$$

Let $\mathsf{LeafStop}_{\mathrm{BU}}$ record those two effective stopping assignments. No declared-stop actor is required in this instance.

The local decomposition rule BU-Dep is now explicit. A downward edge from $\mathcal{K}(C_{\mathrm{BU}})$ is licensed when its child kernel records a nonempty subset of $\mathsf{Prem}_{\mathrm{BU}}$ required by the direct proof and the union of all child-premise records covers $\mathsf{Prem}_{\mathrm{BU}}$. The paired upward reconstruction obligation BU-ND is equally explicit: from C_{vis} and C_{bridge}, the common scoped result Q is fixed by the declared verifier-side premise while the unequal bridge-adequacy assignments remain outside the verifier-visible tuple; therefore that common formal acceptance cannot distinguish those assignments, yielding C_{BU}. Both edges in E_{BU} are labeled BU-Dep, and their two leaves jointly discharge BU-ND. No conditional-introduction step is required because the Claim Kernel stores the proposition's hypotheses in $\mathsf{Prem}_{\mathrm{BU}}$ rather than duplicating them inside C_{BU}.

Therefore every edge is licensed by the declared calculus and

$$\mathrm{Sound}(G_{\mathrm{BU}}; C_{\mathrm{BU}}) = 1. \tag{124}$$

The two leaves satisfy their declared effective stopping predicates, so

$$\mathrm{Proper}(G_{\mathrm{BU}}; C_{\mathrm{BU}}) = 1. \tag{125}$$

Reverse traversal of the two recorded BU-Dep edges under the paired BU-ND reconstruction obligation recovers the root conclusion from the leaf boundary:

$$C_{\mathrm{BU}} \in \mathrm{Rec}_{\Gamma_{\mathrm{BU}}}(L_{\mathrm{BU}}; E_{\mathrm{BU}}), \tag{126}$$

therefore

$$\mathrm{Complete}(G_{\mathrm{BU}}; C_{\mathrm{BU}}) = 1, \qquad \mathrm{Adeq}(G_{\mathrm{BU}}; C_{\mathrm{BU}}) = 1. \tag{127}$$

The required leaf-stopping profile is two effective-stop leaves, zero declared-stop leaves, and no declared-stop actor.

The adequacy presentation is itself recorded rather than left as a dangling reference:

RECEIPT ID: R-BU-ADEQ-001
RECEIPT TYPE: structural_adequacy
CARRIES: leaf_stop_profile
STATUS: ADEQUATE — FORMAL STRUCTURAL JURISDICTION ONLY
KERNEL: $\mathscr{K}(C_{\mathrm{BU}})$
GRAPH: G_{BU}, with the two displayed BU-Dep edges
CHECKED: Sound = Proper = Complete = Adeq = 1 under the displayed definitions and local rules.
LEAF STOP PROFILE: 2 effective / 0 declared.
DECLARED-STOP ACTORS: none.
JURISDICTION: formal structural adequacy of this decomposition.
NOT CHECKED: empirical adequacy of either bridge, uniqueness of this decomposition, or any broader ontological interpretation.
RESIDUE: alternative decompositions and external bridge assessments remain persistently challenge-able.

This adequacy result certifies the declared decomposition and reconstruction of Proposition 3.2 under the stated formal boundary. It does not establish the empirical adequacy of either bridge represented by A_{in}, and it does not require competing adequate decompositions to be impossible. The worked graph therefore demonstrates the intended chain from scoped claim to Claim Kernel, declared leaf boundary, recorded reconstruction, adequacy result, and scoped receipt without promoting structural adequacy into external bridge adjudication.

Self-application status. The adequacy determination in this worked example is itself a methodological claim made by the author under the declared formal jurisdiction. Receipt R-BU-ADEQ-001 records that determination rather than rendering it unchallengeable. Challenges to the decomposition, edge licensing, stopping boundary, or reconstruction remain admissible under the persistent-challenge rule and are to be registered by type and disposition.

26. Discussion: Meaning as a Typed Translation Problem

The method suggests a useful reframing of a familiar difficulty. Formalization is itself a translation.

A conceptual claim C becomes a formal object $\phi(C)$ only after choices about vocabulary, identity, admissible transformation, quantification, and scope. A formal result then becomes a scientific or philosophical conclusion only after another translation.

Thus

$$C \xrightarrow{\tau_{\text{formalize}}} \phi(C) \xrightarrow{\text{adjudicate}} Q \xrightarrow{\tau_{\text{interpret}}} I. \tag{128}$$

Neither translation is necessarily lossless.

For a declared partial translation $\tau_{A\to B} : S_A \rightharpoonup S_B$, define the typed directional translation residue by

$$\mathrm{Res}_{A\to B}(\tau_{A\to B}) := S_B \setminus \mathrm{Im}(\tau_{A\to B}), \qquad y \in \mathrm{Res}_{A\to B}(\tau_{A\to B}) \iff y \in S_B \ \wedge \ y \notin \mathrm{Im}(\tau_{A\to B}). \tag{129}$$

The indices identify the declared source and target descriptions or registries; they do not by themselves assert that either description exhausts a physical system. Thus the formalization and interpretation seams above each instantiate $\mathrm{Res}_{A\to B}(\tau_{A\to B})$ with the source and target description domains declared by the corresponding seam: the conceptual-to-formal domain for $\tau_{\text{formalize}}$, and the scoped-result-to-interpretation domain for $\tau_{\text{interpret}}$. Membership in such a residue is an exact statement about the declared translation and target structure; any claim that the residual element corresponds to an objectively missing property of reality requires a further bridge. The purpose of the audit is not to eliminate residue. It is to expose and type it.

This perspective explains why disputes can persist even when participants agree on equations. They may disagree about the formalization bridge, the correspondence bridge, the reference state, or the explanatory role assigned to an invariant. Conversely, two apparently opposed interpretations may share a large formal invariant.

The method therefore treats disagreement itself as decomposable.

26.1 Semantic compression and the pharmakon of formalization

Formalization is productive because it compresses. A long history of measurement choices, equivalence rules, approximations, reference states, and correspondence assumptions can become a compact symbol that later reasoning manipulates efficiently. The same compression can hide the route by which the symbol acquired its meaning.

This is the paper's recurring warning about nominalized explanations. A sequence such as

$$\text{residue} \to \text{parameter} \to \text{degree of freedom} \to \text{entity} \to \text{cause} \tag{130}$$

may contain well-supported transitions, unsupported transitions, or merely definitional transitions. Smooth prose does not make their warrants identical. The appropriate response is not to ban the nouns but to preserve enough provenance to reopen the compressed chain when a conclusion depends on it.

The same principle explains why a formal verifier can certify evidentiary accuracy only for the scoped formal object supplied to it. Lean, Coq, Isabelle, or another checker can verify a derivation from formalized premises; it does not independently adjudicate the interpretation that selected those premises, the empirical encounter from which a registration was produced, or the ontology later attributed to the formal symbols. Those remain separately typed bridge or registration questions unless formalized in turn.

27. Conclusion

Rigorous reasoning requires more than checking the middle of an argument.

A datum is registered through a measurement or representational system. A problem is constructed from that datum using primitives, references, equivalences, and explanatory commitments. A formal or empirical traversal produces a scoped result. That result is then translated into broader meaning.

The full chain is

$$\boxed{X \xrightarrow{\mathcal{M}} D \xrightarrow[\text{audit}]{B_{\text{in}}} P \xrightarrow[\text{verify}]{F} Q \xrightarrow[\text{audit}]{B_{\text{out}}} I}. \tag{131}$$

The central rule is correspondingly simple:

$$\boxed{\text{Every arrow requires a scoped receipt.}} \tag{132}$$

The constructive complement is equally important:

$$\boxed{\text{decompose downward to a declared primitive seam} \;\rightleftarrows\; \text{construct upward through checkable operations}}. \tag{133}$$

A proof-bearing result can therefore be distinguished from a persuasive conclusion, a black-box verdict from a reconstructively accessible certificate, a failed search from a theorem of unreachability, and a finite witness from the theorem that governs its continuation.

This rule protects formal rigor without pretending that formalization exhausts meaning. It protects empirical evidence without turning a measurement convention into an ontology. It permits philosophical interpretation without attributing that interpretation to a theorem that never contained it. It permits canonical conventions to retain their demonstrated robustness while leaving their scope explicit. And it gives independent or unconventional work a common adjudication surface rather than either an exemption from rigor or a presumption of error.

The method is recursive because every unresolved bridge can become another Claim Kernel. It is finite in application because every receipt declares its primitive boundary. It is neutral in aspiration because canonical and challenger claims receive the same type discipline.

The intended result is not a machine that decides every question. It is a filter that makes clearer which question has actually been asked, which part has actually been checked, and which additional meaning has been supplied by the path between them.

Appendix A — Claim Kernel Schema

A machine-readable implementation may represent a kernel using fields equivalent to:

```
claim_id: string
claim_text: string
claim_type:
  - formal
  - empirical
  - semantic
  - methodological
  - bridge
  - zero_reference_limit
premises: []
definitions: []
dependencies: []
admissible_rules: []
jurisdiction: []
measurement_registry: {}
problem_bridge: {}
interpretive_bridge: {}
uncertainty: {}
verification_status: untested | partial | verified | falsified | out_of_scope
residue: []
findings:
  - finding_id: string
    target: string
    finding_type: string
    asserted_obstruction: string
    novelty_relation:
      novel | materially_equivalent | refinement
    revision_relation:
      revision_introduced | dependency_exposed
      | previously_undetected | unchanged_unknown
    related_findings: []
    assessments:
      - actor: string
        jurisdiction: string
        affected_requirement: string | null
        scope: string
        materiality: string
        blocking_status:
          blocking | non_blocking | contested | unresolved
        disposition:
          corrected | blocking | contested | accepted_limitation
          | unresolved | externally_adjudicated
          | out_of_jurisdiction | non_blocking
        rationale: string
receipts:
  - receipt_id: string
    receipt_type: scoped_verification | structural_adequacy | other_scoped
    artifact_ref: string
    carries: [leaf_stop_profile | declared_stop_actors | other_declared]
```

The schema deliberately separates `verification_status` from document-level reputation or publication status. A finding's `novelty_relation` describes its relation to prior registered findings, while `revision_relation` records its provenance with respect to the present revision; the two axes may vary independently. Within `novelty_relation`, `materially_equivalent` includes a restatement or reformulation that preserves the same asserted obstruction and introduces no materially distinct dependency, counterexample, premise dispute, or scope conflict. A `refinement` adds material specificity to an existing obstruction without becoming a distinct obstruction. This removes a redundant `restatement` enum value while retaining linkage to the prior wording through `related_findings`. Requirement, scope, jurisdiction, materiality, blocking status, and disposition are assessment-level fields because different auditors may legitimately disagree about them while referring to the same finding. Any assessment marked `blocking` must identify a non-null `affected_requirement`. Claim-Kernel serialization does not purport to contain every graph or receipt object: decomposition graphs, LeafStop metadata, Zero-Audit actor records, and adequacy receipts are linked supplemental artifacts referenced through `dependencies`, `residue`, or typed `receipts` links rather than silently added to the eight-tuple. A `structural_adequacy` link that presents Adeq $= 1$ is valid only when its linked artifact declares `leaf_stop_profile` in `carries`; if any leaf stopped by declaration, the link must also declare `declared_stop_actors`. The graph therefore remains a supplemental artifact while the normative LeafStop surfacing obligation becomes mechanically checkable at the receipt-link boundary.

The schema is an implementation representation, not a second definition of the Claim Kernel. Its declared presentation map σ_A instantiates Definition 5.1 as follows:

$$
\begin{aligned}
\sigma_A(C) &= \texttt{claim\_text}, & \sigma_A(T) &= \texttt{claim\_type}, \\
\sigma_A(\mathsf{Prem}) &= \texttt{premises}, & \sigma_A(\mathsf{Def}) &= (\texttt{definitions}, \texttt{dependencies}), \\
\sigma_A(\Gamma) &= \texttt{admissible\_rules}, & \sigma_A(J) &= \texttt{jurisdiction}, \\
\sigma_A(\mathcal{X}) &= (\texttt{problem\_bridge}, \texttt{interpretive\_bridge}), & \sigma_A(\rho) &= \texttt{residue}.
\end{aligned}
\tag{134}
$$

The fields `claim_id`, `measurement_registry`, `uncertainty`, `verification_status`, `findings`, and `receipts` are supplemental implementation metadata. The `findings` records instantiate the disposition vocabulary of the global meta-rule so that discovery, novelty relation, materiality, blocking status, disposition, actor, and challenge provenance can be serialized rather than remaining prose-only governance. A finding is stored once while plural `assessments` preserve attributable disagreement among auditors; novelty classification does not determine blocking status. They do not silently enlarge the eight-tuple $\mathcal{K}(C)$; when they become substantive claims, they are represented through the appropriate bridge, receipt, registry, or linked kernel.

Appendix B — Problem-Formation Audit Template

For each problem P:

1. **Datum:** What is actually registered or stipulated?
2. **Measurement registry:** How was it registered, calibrated, enumerated, and bounded?
3. **Reference:** Relative to what origin, null, equilibrium, symmetry, or ideal is discrepancy defined?
4. **Identity:** What counts as the same object or state?
5. **Ordering:** What temporal, causal, or derivational order is assumed?
6. **Symmetry:** Which invariances are primitive, expected, derived, or approximate?
7. **Problem bridge:** Which assumptions transform the datum into the stated problem?
8. **Alternatives:** Does another admissible registry preserve the datum while changing the problem?
9. **Invariant:** What survives the translation?
10. **Residue:** What fails to translate automatically?

Appendix C — Zero Audit Template

For every substantive use of 0:

1. Identify its zero denotation and role labels from Definition 9.1.
2. Identify the operation that produced or selected it.
3. Determine whether it is exact, conventional, limiting, or uncertainty-bounded.
4. If empirical, record the detector model and sensitivity.
5. If a limit, distinguish every finite station from the endpoint.
6. If a state label, do not infer zero physical content without a bridge.
7. If an expectation value, distinguish it from pointwise value and variance.
8. If a renormalization or reference zero, test which observables are invariant under shifts.
9. If an ontological absence claim appears, identify the bridge that licenses it.
10. Record any Zero-Type Error or Reference Reification Error.

Appendix D — Scoped Receipt Example

```
RECEIPT ID: R-F-017
KERNEL: K-042
JURISDICTION: Formal / real analysis
PREMISES: A1-A6
CHECKED: Proposition Q follows from A1-A6
STATUS: VERIFIED
UNCERTAINTY / CONDITIONALITY: none beyond declared premises
NOT CHECKED:
  - physical identification B-11
  - calibration assumption M-04
  - ontological interpretation S-09
RESIDUE:
  - uniqueness of physical correspondence
```

```
DEPENDENCIES:
  - R-D-003 definitions
```

This human-readable block is related to the generalized receipt by a declared presentation map σ_D: `RECEIPT ID`, `KERNEL`, `JURISDICTION`, `PREMISES`, `CHECKED`, `UNCERTAINTY / CONDITIONALITY`, `RESIDUE`, and `DEPENDENCIES` present respectively the corresponding fields of $\mathfrak{R}$ in Equation (54). `STATUS` is a presentation-level summary derived from the scoped check, while `NOT CHECKED` explicitly lists out-of-jurisdiction content and is not an additional tuple component. Structural-adequacy receipts use the same generalized-receipt presentation and may additionally display the presentation-level fields `RECEIPT TYPE`, `CARRIES`, `GRAPH`, `LEAF STOP PROFILE`, and `DECLARED-STOP ACTORS`. These fields do not enlarge $\mathfrak{R}$: they identify the supplemental artifact type and expose the graph/stopping metadata required by Definition 5.2. For a positive `structural_adequacy` presentation, `GRAPH` identifies the audited decomposition, `LEAF STOP PROFILE` reports the effective versus declared leaves, and `DECLARED-STOP ACTORS` reports the responsible actors whenever declared stops occur; `CARRIES` announces which of those supplemental obligations the linked artifact contains.

A public-facing badge should never display only `VERIFIED`; it should display or link to the jurisdiction and scope.

Appendix E — Research Program

The method suggests several immediate research tasks:

1. Build an annotated corpus of papers containing mixed formal, empirical, and interpretive claims.
2. Measure inter-auditor agreement on Claim Kernel decomposition.
3. Develop automatic bridge detection using linguistic and dependency cues.
4. Integrate formal kernels with theorem provers and numerical checkers.
5. Implement Zero Audits for physics papers using null, vacuum, baseline, and limiting language.
6. Compare canonical and independent submissions under blinded formal routing.
7. Develop receipt-composition rules and cryptographically stable machine-readable receipts.
8. Test whether Problem-Formation Audits reveal genuinely different problems while preserving agreed observations.
9. Measure whether the method reduces false attribution of interpretive conclusions to formal results.
10. Subject this method itself to the same decomposition and routing protocol.

Appendix F — Global Notation Register

The register is normative for this paper. A symbol has one principal type and role unless an indexed variant is explicitly declared.

Symbol	Type / role
X	source observational record or target relation prior to registration
$\mathcal{M}$	measurement/registration registry
χ	declared coupling or observable in a measurement registry
Δ	selected increment or comparison structure inside $\mathcal{M}$; distinct from the registry-relative displacement $\Delta_{\mathcal{G}}(D)$
$q_{\mathcal{M}}$ (or q when the registry is explicit)	registration/readout map from the source record to the reported datum or estimate
κ_N	finite quotient/readout map from a cyclic state domain to N retained bins in Section 4.3; distinct from the scoped-result symbol Q
$\mathcal{N}$	numerical representation used by a measurement registry
ϵ	resolution or uncertainty structure carried by a measurement registry $\mathcal{M}$

Symbol	Type / role
$\epsilon_{\mathcal{M}}$	instantiated measurement-resolution bound under registry $\mathcal{M}$; not a new registry component
U_{cal}	unit/calibration convention
D	registered datum or estimate
B_{in}	upstream problem-forming bridge
P	problem submitted for adjudication
Obj	submitted proof-bearing object
Ver	verifier for a proof-bearing object
F	formal, computational, or empirical traversal
Q	scoped result
B_{out}	downstream interpretive/correspondence bridge
I	broader interpretive conclusion reached through B_{out}
H	human or independent machine auditor; when used in a declared stopping predicate, the actor declaring that boundary
H in Section 13	local Hamiltonian operator in the quantum-vacuum worked case; distinct from H as the human/machine auditor or declared stopping actor elsewhere in the paper
E_n, E_0 in Section 13	local Hamiltonian energy eigenvalues/levels; distinct from the routing type tag E and graph-edge sets E_C, E_G
C	scoped claim
$\mathsf{Ch}, \mathsf{Ch}_i$	registration chain of Definition 3.1 and its indexed witnesses in Proposition 3.2
$C_{\mathrm{BU}}, C_{\mathrm{vis}}, C_{\mathrm{bridge}}, \mathsf{Prem}_{\mathrm{BU}}, \mathsf{Def}_{\mathrm{BU}}, \Gamma_{\mathrm{BU}}, J_{\mathrm{BU}}, \mathcal{X}_{\mathrm{BU}}, \rho_{\mathrm{BU}}$	local scoped conclusion, leaf claims, and indexed Claim-Kernel fields in the worked self-application of Section 25; no new principal types are introduced
$G_{\mathrm{BU}}, V_{\mathrm{BU}}, E_{\mathrm{BU}}, L_{\mathrm{BU}}, \mathsf{LeafStop}_{\mathrm{BU}}$	local kernel-decomposition graph, vertex/edge sets, leaf boundary, and leaf-stopping assignment in the worked decomposition of Section 25.1
$A_{J_{\mathrm{BU}}}$	independently declared decidable set of premise forms admitted by the finite Γ_{BU} grammar in Section 25.1; the boundary is fixed before leaf checks and each finite leaf premise set is tested by literal membership/subset inclusion
BU-Dep, BU-ND	paired local rules in the worked Proposition 3.2 decomposition: BU-Dep licenses downward premise-dependency edges covering $\mathsf{Prem}_{\mathrm{BU}}$, and BU-ND reconstructs C_{BU} from the two declared leaves under Γ_{BU}
$X_n, X_{n+1}, \widetilde{X}_1, \widetilde{X}_2$	lower-resolution, lifted, and competing extended state spaces in Section 14.1; model-description spaces distinct from source-record X
$\tilde{x}_1, \tilde{x}_2$	states in the competing extended spaces of Section 14.1
π_1, π_2	projections from $\widetilde{X}_1, \widetilde{X}_2$ to their common lower-resolution description X_n
r, Y_r	discriminating observable and declared common codomain in Section 14.1, with $r : \widetilde{X}_1 \sqcup \widetilde{X}_2 \to Y_r$
C_i, B_i	subclaims and subordinate bridges produced by recursive decomposition (§10)
$C^{(k)}$	constructive state in a finite constructive trace

Symbol	Type / role
$\phi(C)$	formalized image of a conceptual claim under $\tau_{\text{formalize}}$; distinct from the traversal F
g	local recursion/fixed-point map used only in Section 20.2; distinct from the traversal F and the type tag F
$\mathcal{K}(C)$	Claim Kernel
Prem	premise set of a claim
Π_{prim}	primitive set of a constructive grammar
β_{pc}	partial premise-to-primitive seam/compilation map
Def	definitions and typed dependencies
Γ	admissible grammar/rule set
$J, \mathcal{J}$	jurisdiction / set of adjudicator classes
$\mathscr{R}_{\text{route}}$	partial, set-valued routing relation
ρ	unresolved claim-kernel residue only
ρ_{rec}	unresolved residue in a generalized receipt
$\text{Res}_n^{\text{mod}}$	model-closure residual for an indexed restricted or enlarged description in Section 14.1; distinct from Claim-Kernel residue ρ, receipt residue ρ_{rec}, and typed translation residue $\text{Res}_{A\to B}$
$\Delta_{\mathcal{G}}(D)$	reference-relative displacement in registry $\mathcal{G}$
$\mathfrak{R}$	scoped verification receipt
id	receipt identifier field in the generalized receipt tuple; an administrative identifier, not an inference rule or identity predicate
$\text{R–BU–001}, \text{R–BU–ADEQ–001}$	local human-readable receipt identifiers for the Proposition 3.2 formal-result receipt and structural-adequacy receipt, respectively
U_{unc}	uncertainty or conditionality recorded in a receipt
$\tau_{A\to B}, \tau_{ij}$	partial translation maps between declared source and target descriptions or registries
S_A, S_B	declared source and target structures for a typed translation residue; description-level domains/codomains, not claims of exhaustive physical ontology
$\text{Res}_{A\to B}(\tau_{A\to B})$	typed directional translation residue $S_B \setminus \text{Im}(\tau_{A\to B})$; indexed forms such as $\text{Res}_{i\to j}$ and $\text{Res}_{\text{std}\to\Lambda}$ are instances of this family
$\mathfrak{S}_i$	named target structures for particular translation comparisons; each is an instance of the general target structure S_B when it occupies the target role (for example $S_B = \mathfrak{S}_\Lambda$ in Section 12.2)
$\mathcal{S}_{\text{sym}}$	symmetry commitments in a problem-forming registry
$\mathcal{O}_{\text{ord}}$	ordering assumptions in a problem-forming registry
$\mathcal{O}_{\text{dyn}}$	explicitly declared dynamical specialization of $\mathcal{O}_{\text{ord}}$ used in Section 12
$\mathscr{O}_{\text{oracle}}$	opaque oracle verdict relation
$V_G, E_G \,/\, V_C, E_C$	vertex and edge sets of a document claim graph and of a finite rooted directed acyclic Claim-Kernel decomposition graph; in G_C, $\mathcal{K}(C)$ is the in-degree-zero root, every other vertex is reachable from it, and edges point toward typed dependencies, subclaims, premise-bearing kernels, or bridge obligations

Symbol	Type / role
Prim	mathematical/problem-forming primitive commitments when a prose category is required
$\mathcal{C}_{\mathrm{corr}}$	calibration and correspondence assumptions
$\mathcal{Z}$	reference/zero commitments
$\mathcal{E}_{\mathrm{enum}}$	enumeration/estimation procedure in a measurement registry
$\mathcal{E}_{\mathrm{eqv}}$	identity and equivalence conventions
Φ	problem-forming functional
$\mathcal{G}$	problem-generating registry; indexed variants identify the registry under comparison
$\mathrm{Rec}_{\Gamma_C}(L_C; E_C)$	typed kernel-decomposition recoverability operator: reconstructs from the declared leaf boundary through the recorded edge structure under the reconstruction obligations of Γ_C; non-leaf vertices are not independent starting premises
$\mathrm{Rec}_{\Gamma}(\mathsf{Tgt})$	positional/target-structure recoverability operator used outside Definition 5.2; distinct in argument type from kernel-decomposition reconstruction
$\mathrm{Leaves}(G_C), L_C$	leaf-boundary operator and resulting set of out-degree-zero Claim Kernels in a finite DAG G_C
Supp_{Γ}	supplementary residue relative to recoverable content
Dep	dependency operator
Sound	edge-licensing predicate: every decomposition edge in E_C must be licensed by the declared typed decomposition/bridge calculus Γ_C
Complete	decomposition-completeness predicate: recovery of root claim C from L_C along E_C under the declared reconstruction obligations of Γ_C
Proper	leaf-stopping condition relative to $\mathsf{LeafStop}_C$ and Definition 10.1
$\mathsf{LeafStop}_C$	declared leaf-stopping assignment carried as metadata of a kernel decomposition graph; maps each leaf kernel to jurisdiction, stop mode, and actor when required
$\mathrm{Stop}_J^{\mathrm{eff}}, \mathrm{Stop}_J^{\mathrm{decl}}$	effective and declared stopping predicates of Definition 10.1, applied to Claim Kernels
Λ as a subscript/target label	Lambda registry label in $\mathcal{G}_\Lambda$, $\mathfrak{S}_\Lambda$, and $\tau_{\mathrm{std}\to\Lambda}$; not a leaf-stopping assignment
Adeq	sound, complete, and proper decomposition predicate; any presented positive result carries its $\mathsf{LeafStop}_C$ profile so effective and declared stopping remain distinguishable
RA	reconstructive accessibility predicate
$\mathsf{ZeroClass}, Z_{\mathrm{den}}, Z_{\mathrm{role}}$	zero-classification function and its denotation/role component classifiers
$\mathrm{Admissible}_S$	semantic/type-S admitted relation on the finite adjudication domain $\mathcal{D}_S$ in Section 9; actor-supplied and retained as inspectable residue, while product-order minimization is formal only relative to those registered judgments

Symbol	Type / role
$\mathcal{D}_S$	declared finite domain of occurrence-label pairs for which the Section 9 actor has recorded explicit admitted/rejected semantic judgments; unadjudicated pairs remain outside the formal minimization domain rather than defaulting to false
$\mathcal{O}_{\mathrm{act}}$	declared finite set of inferentially active zero-like occurrences over which the joint minimal-cover condition in Section 9 is assessed
$\widehat{\mathcal{Z}}_{\mathrm{den}}, \widehat{\mathcal{Z}}_{\mathrm{role}}$	candidate denotation/role accounting pair for $\mathcal{O}_{\mathrm{act}}$; primed forms denote competing candidate pairs in the joint product-order minimality test
$\mathcal{Z}_{\mathrm{den}}, \mathcal{Z}_{\mathrm{role}}$	declared denotation-label and role-label vocabularies used by the zero classifier; distinct from $\mathcal{Z}$, the reference/zero component of B_{in}
$\mathrm{Set}(\mathcal{J})$	set of admissible adjudicator classes; used instead of $\mathcal{P}(\mathcal{J})$ to avoid collision with primitive notation
$\mathcal{T}$	set of provisional claim-type routing tags $\{F, E, S, M, B, Z\}$
T	provisional claim-type field, with $T \in \mathcal{T}$
Tgt	target structure in positional-recoverability statements
$A_{\mathrm{in}}, A_F, A_J, A_{\mathrm{rec}}$	typed adequacy/acceptance fields: input-bridge adequacy, formal-receipt premises, jurisdiction boundary, and generalized-receipt assumptions
$K_F^{\checkmark}, K_{\mathrm{rec}}^{\checkmark}$	verified-content fields in formal and generalized receipts
$\mathcal{X}$	bridges from a Claim Kernel to other kernels
Σ	dependencies of a generalized receipt on other receipts
$\mathfrak{D}$	downward claim-decomposition operator
$\eta_B, n_B, n_b, n_{\bar{b}}, n_\gamma$	local baryon/photon ratio and number-density symbols in Section 12; physics-local quantities, not routing tags
$\hat{\eta}_B, X_{\mathrm{BBN}}, \mathcal{M}_{\mathrm{BBN}}, P_{\mathrm{bar}}$	local Section 12 registered estimate/source/measurement-registry/problem symbols for the baryon-asymmetry example
c_0	scalar constant used for the Hamiltonian energy-origin shift
u_j	advertised-indispensable primitive tested for dependency in Section 15.2
γ_k	indexed constructive-trace operation in Definition 15.1, with $\gamma_k \in \Gamma$
Macro	admitted macro library in constructive dependency tests
Cert	imported-certificate dependency set in constructive dependency tests
$\mathsf{PS}_1, \ldots, \mathsf{PS}_4$	proof-status relations: verifier acceptance, valid expansion, reconstructive accessibility, and interpretive assent
$\mathrm{Cmp}_{\mathrm{Adeq},C}$	pairwise adequacy-comparison predicate under the declared comparison map μ_{cmp}
μ_{cmp}	declared comparison map used only by $\mathrm{Cmp}_{\mathrm{Adeq},C}$; distinct from the measurement coupling/observable χ
E_{cross}	unflagged invalid cross-type inference rate on an eligible adjudicated benchmark; superscripts baseline and method denote the compared procedures

Symbol	Type / role
F_{flag}	false-flag rate over adjudicated valid cross-type transitions; superscripts baseline and method denote the compared procedures
$N_{\text{invalid,adjudicated}}, N_{\text{invalid,unflagged}}$	invalid-transition benchmark counts used to define E_{cross} in Section 23.6
$N_{\text{valid,adjudicated}}, N_{\text{valid,flagged}}$	valid-transition benchmark counts used to define F_{flag} in Section 23.6
δ_{cross}	preregistered minimum improvement margin for E_{cross}
$\varepsilon_{\text{flag}}$	preregistered maximum admissible false-flag rate
δ_{flag}	preregistered maximum baseline-relative increase permitted in the false-flag rate
`finding.novelty_relation`	relation of a finding to prior registered findings: novel, materially equivalent, or refinement; materially equivalent includes non-novel restatements/reformulations of the same obstruction; independent of revision provenance, materiality, and blocking status
`finding.revision_relation`	revision provenance of a finding: revision introduced, dependency exposed, previously undetected in unchanged material, or unchanged/unknown
`receipt.receipt_type`, `receipt.carries`	typed supplemental-receipt linkage; a positive structural-adequacy receipt must carry a leaf-stop profile and, when applicable, declared-stop actors
`GRAPH`, `LEAF STOP PROFILE`, `DECLARED-STOP ACTORS`	structural-adequacy receipt presentation fields declared in Appendix D; they expose the audited graph and Definition 5.2 stopping metadata without enlarging the generalized receipt tuple
`finding.assessments[]`	plural attributable auditor assessments; requirement, scope, jurisdiction, materiality, blocking status, disposition, and rationale remain assessment-level rather than being collapsed into the finding itself
`finding.assessments[].disposition`	machine-readable disposition vocabulary implementing Section 2.1: corrected, blocking, contested, accepted limitation, unresolved, externally adjudicated, out of jurisdiction, or non-blocking
σ_A, σ_D	declared presentation maps from the formal Claim Kernel and generalized receipt to Appendices A and D
$\mathcal{E}_O, d_O$	local empirical-encounter symbol and the shareable registration produced from it in Section 4.2
$D_R, \mathcal{L}_R, \sim_R$	registry discrimination map, retained label space, and induced registry-relative equivalence in Section 4.3
θ	local cyclic coordinate in the variable example; its finite quotient/readout map is κ_N
Real_R	registry-indexed reality predicate whose prerequisites must be declared
ξ, L, π	local closure degree of freedom, lift, and projection in Section 14.1; the lift map L is distinct from the kernel-graph leaf boundary L_C
TOE	Theory of Everything; routing label used for TOE Arena/formal traversal infrastructure

The symbols $\{F, E, S, M, B, Z\}$ in the provisional type set are kernel-type tags, not mathematical variables; their local tag role is distinct from traversal, bridge, or registry symbols using similar glyphs.

Premises and primitives are deliberately distinct. A premise accepted for a claim need not already be a primitive operation of the constructive grammar. The seam β_{pc} makes that conversion auditable rather than implicit.

Resolution notation is likewise typed. ϵ denotes the registry-level resolution/uncertainty structure; $\epsilon_{\mathcal{M}}$ is its instantiated measurement bound. Indexed $\epsilon_1, \epsilon_2, \ldots$ are declared instances of that same resolution quantity. $\varepsilon_{\mathrm{flag}}$ is a separate preregistered false-flag threshold in Section 23.6, not an indexed instance of ϵ.

Kernel-graph orientation is fixed by Definition 5.2: edges run from a root/dependent kernel toward typed grounds, subclaims, premise-bearing kernels, or bridge obligations; L_C is therefore the out-degree-zero stopping boundary. Sound certifies licensed downward edges; Complete certifies upward reconstruction from that boundary along the same recorded edge structure.

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Likewise, this paper's use of conventional baryogenesis and quantum-vacuum examples does not allege defects in those fields. The companion Lambda paper is treated as an application of the present claim-routing method, not as a premise required for accepting the method. The examples are used to demonstrate claim typing and jurisdiction: a successful formal or empirical result retains its status while the assumptions that form the problem and the bridges that extend the result remain separately auditable.

CERTIFIED APPENDIX

Executable Claim Kernel, Lean-Checked Receipt, Artifact Identity, and Verification Boundary

A self-application of Before the Proof, Proposition 3.2 / C_{BU}

STATUS: LEAN-CHECKED - DECLARED FORMAL JURISDICTION ONLY

1. Purpose and scope

This appendix is an executable exemplar of the paper's claim-routing method. It certifies one encoded, scoped dependency result. It is **not** a blanket certification of the paper, its empirical examples, its interpretations, or the fidelity of every translation into formal language.

CERTIFIED TARGET

The target is Proposition 3.2 / C_{BU}: acceptance of a shared verifier-visible formal traversal underdetermines a declared adequacy property of upstream bridges when that property is omitted from the verifier input.

FROZEN PARENT ARTIFACT

Before_the_Proof_V1_PUBLICATION_CLOSURE_FINAL_R2.pdf
42 pages | 268,744 bytes | SHA-256 `224888da1fd46a7f53c13ec006035bf9e3b6fa617bd244bc1ad79fae9a15d20a`

The parent PDF was not re-typeset. Its pages were cloned into the certified edition before these appendix pages were appended.

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2. Selected Claim Kernel

Paper claim C_{BU}

SCOPED CONCLUSION

Acceptance of the shared verifier-visible formal traversal underdetermines the declared adequacy property of the two upstream bridges.

$Prem_{BU}$

The paper states that two well-formed registration chains share formal traversal F and the verifier-visible tuple (P, $Prem_F$, $Gamma_F$); the scoped verification relation depends only on that tuple and yields common accepted result Q; and $A_{in}(B_{in}^{(1)}) = 1$ while $A_{in}(B_{in}^{(2)}) = 0$.

The encoded premises are:

- **hInput**: both chains expose equal verifier-visible inputs.
- **hResult1, hResult2**: each stored result is produced by the same verifier from its visible input.
- **hAdeq1, hAdeq2**: the upstream bridges receive opposite adequacy assignments.
- **Type separation**: verifier input and upstream bridge are stored as distinct fields.

Scoped inference

Substitution through hInput gives equality of verifier outputs. The stored-result equations then give c1.result = c2.result. The adequacy disagreement remains a declared premise on the separate bridge fields. The resulting witness therefore shows that the common verifier-side result does not determine the bridge adequacy assignment.

SEMANTIC COMPARISON FINDING

No meaning-changing correction was required. The theorem is a faithful **formal surrogate** for the dependency-level target, but it is intentionally not a literal encoding of the paper's entire nine-component registration chain, a real-world acceptance procedure, or empirical bridge adequacy. Those limits remain explicit in the correspondence table and receipt.

3. Formalization seam

The seam is directional: natural-language objects are mapped to Lean objects for a declared purpose. Lean checks the theorem after that mapping; it does not independently prove that the mapping exhausts the source meaning.

Paper object	Lean object	Classification	Audit note
C_BU	two_chain_bridge_underdeterminatio n	Formal surrogate	Encodes same visible result together with opposite bridge-adequacy assignments.
Prem_BU	hInput; hResult1; hResult2; hAdeq1; hAdeq2	Formal surrogate	Dependency premises are explicit; full-chain well-formedness is not represented.
Verifier-visible tuple (P, Prem_F, Gamma_F)	VerifierInput.problem / .premises / .grammar; instantiated as Input	Directly encoded	The theorem is polymorphic, but the intended instantiation uses this three-field structure.
Traversal F / scoped verifier	verify : Verifier Input Result	Formal surrogate	A deterministic function from declared input to result; no empirical or operational semantics added.
A_in	Adequate : Bridge -> Prop	Directly encoded	Predicate remains on the bridge type, outside the verifier-input fields.
Common result Q	Result; c1.result; c2.result; verify input	Formal surrogate	Result is intentionally uninterpreted; only equality/dependency is checked.
Two registration chains	RegistrationChain verifierInput upstreamBridge result	Formal surrogate	Reduced to fields needed by C_BU; X, M, D, B_out, and I are not encoded.
Concrete bridge truth and meaning	No Lean object	Outside Lean jurisdiction	Empirical adequacy, semantics, causation, ontology, uniqueness, and completeness remain external.

JURISDICTION BOUNDARY

The theorem checks a relation among typed formal objects. It does not certify that a concrete bridge is adequate, that the reduced structure is a complete model of every registration chain, or that the natural-language-to-Lean correspondence is uniquely correct.

4. Exact Lean source - part 1 of 2

Distributed file: **BeforeTheProof_BU_FINAL.lean** | SHA-256 before publication assembly:
`18f47a8677797108da06057b00cefe5829fe9e748912c6a8441fef29d059dc5c`

```
import Lean

/-!
Before the Proof - Certified Appendix exemplar
Target: Proposition 3.2 / C_BU (bridge-underdetermination)

This file formalizes only the dependency-level claim that a verifier whose
result depends solely on its declared verifier-visible input cannot distinguish
an upstream bridge property that is not part of that input.
-/

universe u v w

namespace BeforeTheProof

structure VerifierInput (Problem Premises Grammar : Type u) where
  problem : Problem
  premises : Premises
  grammar : Grammar

structure RegistrationChain
    (Input : Type u) (Bridge : Type v) (Result : Type w) where
  verifierInput : Input
  upstreamBridge : Bridge
  result : Result

/-- A verifier-visible acceptance relation depends only on the declared input. -/
def Verifier (Input : Type u) (Result : Type w) := Input → Result

/--
Formal surrogate for C_BU.
If two chains expose the same verifier input to a verifier, their verifier-side
results are equal, even when a separately declared bridge predicate differs.
Thus equality of the verifier-visible traversal/result does not determine that
upstream bridge predicate.
-/
theorem bridge_underdetermination
    {Input : Type u} {Bridge : Type v} {Result : Type w}
    (verify : Verifier Input Result)
    (b1 b2 : Bridge)
```

4. Exact Lean source - part 2 of 2

Continuation of the same file; no lines are omitted.

```
    (Adequate : Bridge → Prop)
    (hAdeq1 : Adequate b1)
    (hAdeq2 : ¬ Adequate b2)
    (i : Input) :
    verify i = verify i ∧ Adequate b1 ∧ ¬ Adequate b2 := by
  exact ⟨rfl, hAdeq1, hAdeq2⟩

/--
A two-chain presentation closer to Proposition 3.2: once the two chains have
identical verifier-visible input and their stored results are produced by the
same verifier, the results coincide; the bridge-adequacy disagreement remains
an independent premise rather than something certified by that result.
-/
theorem two_chain_bridge_underdetermination
    {Input : Type u} {Bridge : Type v} {Result : Type w}
    (verify : Verifier Input Result)
    (c1 c2 : RegistrationChain Input Bridge Result)
    (hInput : c1.verifierInput = c2.verifierInput)
    (hResult1 : c1.result = verify c1.verifierInput)
    (hResult2 : c2.result = verify c2.verifierInput)
    (Adequate : Bridge → Prop)
    (hAdeq1 : Adequate c1.upstreamBridge)
    (hAdeq2 : ¬ Adequate c2.upstreamBridge) :
    c1.result = c2.result ∧
      Adequate c1.upstreamBridge ∧
      ¬ Adequate c2.upstreamBridge := by
  constructor
  · calc
      c1.result = verify c1.verifierInput := hResult1
      _ = verify c2.verifierInput := congrArg verify hInput
      _ = c2.result := hResult2.symm
  · exact ⟨hAdeq1, hAdeq2⟩

end BeforeTheProof

-- Verification-environment receipt when executed by Lean.
#eval Lean.versionString
```

The sole certification-stage source change was adding `import Lean` so the version constant is available, plus `#eval Lean.versionString` to expose the execution version. The definitions, theorem statements, and proofs supplied by the author were otherwise preserved.

5. Actual Lean execution result

Public service	Lean Web
Service URL	https://live.lean-lang.org/
Verification UTC	2026-08-22T04:04:18.019Z to 2026-08-22T04:04:26.576Z
Project	MathlibDemo
Toolchain	leanprover/lean4:v4.34.0-rc2
Lean.versionString	"4.34.0-rc2"
Mathlib host commit	23a3216f0e5b7f9f5d437409db9f10b34844e6d0 (present in host project; not imported by this source)
Imports	import Lean; no Mathlib import
Primary theorem	BeforeTheProof.two_chain_bridge_underdetermination
Checker completion	Lean fileProgress ended with processing: []
Checker result	ACCEPTED - zero error diagnostics

MACHINE-READABLE EVIDENCE

The companion evidence JSON records the exact submitted source, its pre-run SHA-256, public endpoint, toolchain and package manifest, LSP request/response transcript, final empty processing queue, diagnostics, and checker result. The informational diagnostic `"4.34.0-rc2"` is the output of `#eval Lean.versionString`; severity 3 is informational, not an error.

6. Scoped verification receipt

RECEIPT ID	R-BU-LEAN-001
STATUS	LEAN-CHECKED - DECLARED FORMAL JURISDICTION ONLY
KERNEL	K(C_BU), encoded by BeforeTheProof.two_chain_bridge_underdetermination
JURISDICTION	Formal dependency/metalogical claim over the declared Lean objects.
CHECKED	The encoded conclusion follows from the encoded premises under the encoded definitions and Lean rules.
NOT CHECKED	Empirical truth of premises; semantic fidelity beyond the declared seam; physical interpretation; causal attribution; ontology; model completeness; source attribution; or any claim outside the encoded target.
RESIDUE	All explicitly non-formal or externally adjudicated content remains visible rather than being silently promoted or demoted.

Why this is not a good-versus-bad classification

Machine checkability is a jurisdiction label, not a ranking of importance or truth. The method does not sort claims into good/bad, true/false, orthodox/exotic, or scientific/unscientific categories merely by whether Lean can check them. It identifies which warrant attaches to which claim. A formally verified dependency result retains its force; empirical, semantic, methodological, causal, and ontological questions retain their own adjudication requirements.

7. Inspection, reproducibility, and artifact identity

Independent rerun

- Open **https://live.lean-lang.org/** in a JavaScript-enabled browser.
- Select the **MathlibDemo** project, paste the complete contents of **BeforeTheProof_BU_FINAL.lean**, and wait for processing to finish.
- Confirm there are no severity-1 error diagnostics and that the version output is "4.34.0-rc2" for the recorded run environment.
- Compare the source SHA-256 with the manifest and SHA256SUMS.txt before treating the rerun as the same artifact.

REPRODUCIBILITY CONDITION

Lean Web tracks a moving public toolchain. This receipt identifies the exact environment used for the recorded run. A future rerun on a newer environment is a new check and should report its own toolchain and timestamp, even if the source remains unchanged.

SHA-256 and exact byte identity

SHA-256 identifies the exact byte-state of an artifact. Matching digests provide strong evidence that two byte sequences are identical. A digest does **not** certify truth, logical correctness, empirical adequacy, authorship, priority, ontology, interpretation, or endorsement.

The final PDF cannot contain its own final digest: printing that digest inside the PDF would change the PDF's bytes and therefore change the digest. The certified edition is frozen first; its SHA-256 is then recorded in the external manifest and checksum file. The manifest's own digest is recorded in SHA256SUMS.txt, avoiding a self-reference cycle.

AUTHORSHIP, ATTRIBUTION, AND INDEPENDENCE

Lu Semita / EmergenceByDesign
Independent laboratory notes and research record.

Authorship attribution only; no claim of ownership, invention, discovery, priority, exclusivity, or endorsement. The authorship block records responsibility and provenance for this independent written record; it does not transfer ownership of underlying mathematics, physics, observations, prior art, conventional methods, cited work, or independently developed ideas.